Supplementary Material for "Leakage-Aware Multi-Horizon Benchmarking of Compact Neural PV Forecasts Across Technologies and Sites"

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1 Evidence scope

This supplement reports the frozen evidence supporting the main manuscript. The benchmark comprises three co-located Alice Springs arrays (36 runs, 17 channels) and two separately fitted external facilities, Yulara and NIST Ground (24 runs, seven channels). Each experiment uses four compact neural implementations, seeds 42, 43, and 44, and direct 144-step forecasts at five-minute resolution. Primary H12, H48, H96, and H144 metrics use horizon-specific valid origins. Daily Persistence comparisons instead use an elementwise intersection mask that is valid for the daily lag, labels, Last-value Persistence, and every neural forecast.

2 Input and preprocessing configuration

For Alice Springs, each history has 72 time steps and 17 channels: Performance Ratio; ambient temperature; relative humidity; global and diffuse horizontal radiation; global and diffuse tilted radiation; Active Power; eight corresponding missingness indicators; and one Isolation Forest indicator. The imputer, feature scaler, target scaler, and Isolation Forest are fitted on Train only. No forecast-origin successor value enters the input. Windows remain within chronological splits and within a regular five-minute coordinate.

Table S1: Fixed optimization configuration for all compact neural implementations.
Alt text: The shared optimization configuration fixes the optimizer, learning rate, batch size, epoch budget, early-stopping rule, gradient clipping, H144 validation objective, and three random seeds for all four compact implementations.

Setting	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	256
Maximum epochs	25
Early-stopping patience	5
Minimum improvement	10^{-8}
Gradient clipping	1.0
Checkpoint objective	Validation global masked MSE on H144
Random seeds	42, 43, 44

3 Architecture identity and full layer configuration

The Discrete recurrent decoder combines three temporal-convolution branches (dilations 1, 2, and 4; 24 channels each), a 64-dimensional one-layer Transformer, a learned feature gate, bidirectional GRU attention, and an autoregressive GRUCell trajectory decoder. The Inverted-variate Transformer treats the 17 Alice or seven external channels as tokens, projects each 72-point history to 64 dimensions, applies one four-head Transformer layer, then flattens the encoded tokens into a 144-output linear map. The Joint-patch Transformer forms overlapping joint multivariate patches of length 12 and stride 6, uses one 64-dimensional four-head Transformer layer, and maps the encoded patches to 144 outputs. The Depthwise TCN uses an input projection followed by four depthwise-separable temporal blocks with 64 channels and kernel size 5, then a flattened linear H144 head.

These are project implementations for a controlled mechanism comparison. The inverted model omits the full official iTransformer embedding, normalization, and decoder design; the joint-patch model does not use the channel-independent PatchTST pipeline; and the depthwise model omits the full ModernTCN large-kernel and multi-scale block system. Accordingly, the reported rankings do not rank the official architectures.

4 Metrics and masks

For a fixed valid-point set $\mathcal{I}$ with $n = |\mathcal{I}|$,

$$\mathrm{RMSE} = \sqrt{n^{-1}\sum_{i\in\mathcal{I}}(\hat{y}_i - y_i)^2}, \tag{S1}$$

$$\mathrm{MAE} = n^{-1}\sum_{i\in\mathcal{I}}|\hat{y}_i - y_i|, \tag{S2}$$

$$R^2 = 1 - \frac{\sum_{i\in\mathcal{I}}(\hat{y}_i - y_i)^2}{\sum_{i\in\mathcal{I}}(y_i - \bar{y})^2}, \tag{S3}$$

$$\text{range-nRMSE} = \frac{\mathrm{RMSE}}{y_{\mathrm{max,Train}} - y_{\mathrm{min,Train}}}, \tag{S4}$$

$$\text{RMSE skill} = 1 - \frac{\mathrm{RMSE}_{\mathrm{model}}}{\mathrm{RMSE}_{\mathrm{reference}}}. \tag{S5}$$

The daylight mask is true target Active Power greater than 1% of the corresponding array's Train maximum. It is an evaluation-only mask. Daily Persistence takes each target timestamp's power 24 hours earlier without interpolation. If that point is unavailable, it is removed from every method in that Daily-matched comparison.

5 Alice Springs quantitative evidence

The following tables are generated directly from the frozen `corrected_metrics.csv`; they are not transcribed into this source file. They provide per-seed primary metrics, primary support, complete-H144-origin sensitivity, and matched Daily Persistence results.

Table S2: Primary per-seed RMSE values.

Alt text: Per-seed root mean square errors show how forecast accuracy changes across arrays, scopes, and horizons for the four compact neural implementations. Seed spread reflects initialization variability rather than independent data replication.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.20699	0.21172	0.20282
Sanyo	Full	H12	Inverted-variate	0.26513	0.27584	0.22716
Sanyo	Full	H12	Joint-patch	0.29326	0.24817	0.21943
Sanyo	Full	H12	Depthwise TCN	0.23350	0.17550	0.25439
Sanyo	Daylight	H12	Discrete recurrent	0.29601	0.30675	0.29114
Sanyo	Daylight	H12	Inverted-variate	0.31789	0.32546	0.30911
Sanyo	Daylight	H12	Joint-patch	0.37815	0.32296	0.27677
Sanyo	Daylight	H12	Depthwise TCN	0.31123	0.24041	0.32051
Sanyo	Full	H48	Discrete recurrent	0.64683	0.64400	0.66629
Sanyo	Full	H48	Inverted-variate	0.68041	0.62303	0.62109
Sanyo	Full	H48	Joint-patch	0.66859	0.58335	0.59367
Sanyo	Full	H48	Depthwise TCN	0.67710	0.64485	0.63471
Sanyo	Daylight	H48	Discrete recurrent	0.94975	0.94651	0.97978
Sanyo	Daylight	H48	Inverted-variate	0.95042	0.81287	0.85809
Sanyo	Daylight	H48	Joint-patch	0.95240	0.80115	0.83710
Sanyo	Daylight	H48	Depthwise TCN	0.94231	0.90116	0.87112
Sanyo	Full	H96	Discrete recurrent	0.90840	0.90011	0.96696
Sanyo	Full	H96	Inverted-variate	0.85729	0.80883	0.81027
Sanyo	Full	H96	Joint-patch	0.96456	0.84964	0.81070
Sanyo	Full	H96	Depthwise TCN	0.96364	0.85757	0.88544
Sanyo	Daylight	H96	Discrete recurrent	1.34860	1.33368	1.43813
Sanyo	Daylight	H96	Inverted-variate	1.19297	1.08960	1.11853
Sanyo	Daylight	H96	Joint-patch	1.39986	1.20066	1.14794
Sanyo	Daylight	H96	Depthwise TCN	1.36564	1.19882	1.24128
Sanyo	Full	H144	Discrete recurrent	0.87822	0.82591	0.89962
Sanyo	Full	H144	Inverted-variate	0.78097	0.76385	0.78197
Sanyo	Full	H144	Joint-patch	0.84637	0.78215	0.75005
Sanyo	Full	H144	Depthwise TCN	0.85780	0.79629	0.79624
Sanyo	Daylight	H144	Discrete recurrent	1.31199	1.22879	1.34119
Sanyo	Daylight	H144	Inverted-variate	1.07005	1.02634	1.07535
Sanyo	Daylight	H144	Joint-patch	1.22156	1.10516	1.06103
Sanyo	Daylight	H144	Depthwise TCN	1.20759	1.10868	1.11171
Hanwha	Full	H12	Discrete recurrent	0.19283	0.20984	0.23773
Hanwha	Full	H12	Inverted-variate	0.19326	0.21233	0.21254
Hanwha	Full	H12	Joint-patch	0.25045	0.29434	0.23248
Hanwha	Full	H12	Depthwise TCN	0.17950	0.16762	0.17960
Hanwha	Daylight	H12	Discrete recurrent	0.27411	0.30308	0.34010
Hanwha	Daylight	H12	Inverted-variate	0.25729	0.28560	0.28441
Hanwha	Daylight	H12	Joint-patch	0.31341	0.31390	0.28217
Hanwha	Daylight	H12	Depthwise TCN	0.23343	0.22040	0.23794
Hanwha	Full	H48	Discrete recurrent	0.57998	0.57396	0.63790
Hanwha	Full	H48	Inverted-variate	0.54741	0.58955	0.58646
Hanwha	Full	H48	Joint-patch	0.61852	0.57368	0.56605
Hanwha	Full	H48	Depthwise TCN	0.57787	0.58575	0.59752

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Daylight	H48	Discrete recurrent	0.84718	0.84454	0.93710
Hanwha	Daylight	H48	Inverted-variate	0.78356	0.84091	0.82613
Hanwha	Daylight	H48	Joint-patch	0.88106	0.80086	0.79516
Hanwha	Daylight	H48	Depthwise TCN	0.81332	0.82086	0.84039
Hanwha	Full	H96	Discrete recurrent	0.80824	0.85950	0.90481
Hanwha	Full	H96	Inverted-variate	0.84121	0.85535	0.88964
Hanwha	Full	H96	Joint-patch	0.87878	0.76678	0.78162
Hanwha	Full	H96	Depthwise TCN	0.79720	0.79868	0.81582
Hanwha	Daylight	H96	Discrete recurrent	1.19164	1.27417	1.34138
Hanwha	Daylight	H96	Inverted-variate	1.21196	1.22202	1.26041
Hanwha	Daylight	H96	Joint-patch	1.27278	1.08927	1.11426
Hanwha	Daylight	H96	Depthwise TCN	1.13261	1.12349	1.15324
Hanwha	Full	H144	Discrete recurrent	0.76065	0.78426	0.81177
Hanwha	Full	H144	Inverted-variate	0.78583	0.80309	0.82715
Hanwha	Full	H144	Joint-patch	0.79506	0.71160	0.73154
Hanwha	Full	H144	Depthwise TCN	0.72555	0.75028	0.75361
Hanwha	Daylight	H144	Discrete recurrent	1.12700	1.16621	1.20763
Hanwha	Daylight	H144	Inverted-variate	1.12435	1.13795	1.16060
Hanwha	Daylight	H144	Joint-patch	1.14106	1.00309	1.03867
Hanwha	Daylight	H144	Depthwise TCN	1.01361	1.05240	1.05096
Qcells	Full	H12	Discrete recurrent	0.44262	0.73507	1.59137
Qcells	Full	H12	Inverted-variate	0.30176	0.34935	0.33091
Qcells	Full	H12	Joint-patch	0.52988	0.57476	0.49208
Qcells	Full	H12	Depthwise TCN	0.52598	0.30576	0.41648
Qcells	Daylight	H12	Discrete recurrent	0.64247	1.06714	2.31558
Qcells	Daylight	H12	Inverted-variate	0.42778	0.47506	0.46668
Qcells	Daylight	H12	Joint-patch	0.72858	0.81902	0.69755
Qcells	Daylight	H12	Depthwise TCN	0.75796	0.43460	0.60317
Qcells	Full	H48	Discrete recurrent	0.77406	0.94518	1.83265
Qcells	Full	H48	Inverted-variate	0.79118	0.82091	0.81042
Qcells	Full	H48	Joint-patch	0.93780	1.03256	0.98688
Qcells	Full	H48	Depthwise TCN	1.20653	1.04753	1.04156
Qcells	Daylight	H48	Discrete recurrent	1.12663	1.37313	2.67354
Qcells	Daylight	H48	Inverted-variate	1.14009	1.17767	1.16234
Qcells	Daylight	H48	Joint-patch	1.33819	1.49374	1.42803
Qcells	Daylight	H48	Depthwise TCN	1.75350	1.52143	1.51385
Qcells	Full	H96	Discrete recurrent	1.04609	1.15831	1.55933
Qcells	Full	H96	Inverted-variate	1.15002	1.14625	1.11365
Qcells	Full	H96	Joint-patch	1.18191	1.23303	1.17040
Qcells	Full	H96	Depthwise TCN	1.36166	1.43712	1.27932
Qcells	Daylight	H96	Discrete recurrent	1.54353	1.70080	2.29733
Qcells	Daylight	H96	Inverted-variate	1.67275	1.66186	1.60714
Qcells	Daylight	H96	Joint-patch	1.71311	1.79827	1.70418
Qcells	Daylight	H96	Depthwise TCN	1.99499	2.10755	1.86964
Qcells	Full	H144	Discrete recurrent	0.90764	0.92812	0.88946
Qcells	Full	H144	Inverted-variate	0.84633	0.86845	0.81567
Qcells	Full	H144	Joint-patch	0.87026	0.85503	0.82492
Qcells	Full	H144	Depthwise TCN	0.79967	0.79287	0.77389
Qcells	Daylight	H144	Discrete recurrent	1.36729	1.38531	1.32341

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H144	Inverted-variate	1.23444	1.25633	1.17410
Qcells	Daylight	H144	Joint-patch	1.26299	1.25253	1.20653
Qcells	Daylight	H144	Depthwise TCN	1.17205	1.16000	1.12490

Table S3: Primary per-seed MAE values.

Alt text: Per-seed mean absolute errors provide a less tail-sensitive comparison of the four compact neural implementations across arrays, scopes, and horizons. The pattern complements RMSE by reducing the influence of large residuals.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.13583	0.12116	0.12189
Sanyo	Full	H12	Inverted-variate	0.20606	0.20906	0.15958
Sanyo	Full	H12	Joint-patch	0.21315	0.17464	0.16188
Sanyo	Full	H12	Depthwise TCN	0.17238	0.11426	0.20034
Sanyo	Daylight	H12	Discrete recurrent	0.22632	0.21644	0.20251
Sanyo	Daylight	H12	Inverted-variate	0.24672	0.25630	0.24797
Sanyo	Daylight	H12	Joint-patch	0.29948	0.25203	0.21011
Sanyo	Daylight	H12	Depthwise TCN	0.25914	0.17448	0.25712
Sanyo	Full	H48	Discrete recurrent	0.31052	0.26504	0.28877
Sanyo	Full	H48	Inverted-variate	0.41494	0.40296	0.36613
Sanyo	Full	H48	Joint-patch	0.37397	0.33657	0.32167
Sanyo	Full	H48	Depthwise TCN	0.40982	0.37110	0.37789
Sanyo	Daylight	H48	Discrete recurrent	0.59637	0.52821	0.55362
Sanyo	Daylight	H48	Inverted-variate	0.63958	0.52668	0.58159
Sanyo	Daylight	H48	Joint-patch	0.61831	0.49596	0.51232
Sanyo	Daylight	H48	Depthwise TCN	0.64092	0.58876	0.54797
Sanyo	Full	H96	Discrete recurrent	0.45539	0.42600	0.46402
Sanyo	Full	H96	Inverted-variate	0.54810	0.53406	0.50433
Sanyo	Full	H96	Joint-patch	0.55816	0.51419	0.46833
Sanyo	Full	H96	Depthwise TCN	0.61608	0.52564	0.54538
Sanyo	Daylight	H96	Discrete recurrent	0.92720	0.88733	0.95360
Sanyo	Daylight	H96	Inverted-variate	0.88941	0.80380	0.83193
Sanyo	Daylight	H96	Joint-patch	1.00084	0.85190	0.79454
Sanyo	Daylight	H96	Depthwise TCN	1.05117	0.88114	0.89613
Sanyo	Full	H144	Discrete recurrent	0.45801	0.40717	0.46249
Sanyo	Full	H144	Inverted-variate	0.50741	0.51151	0.50399
Sanyo	Full	H144	Joint-patch	0.49204	0.47784	0.44669
Sanyo	Full	H144	Depthwise TCN	0.56495	0.51928	0.51794
Sanyo	Daylight	H144	Discrete recurrent	0.96352	0.86876	0.97403
Sanyo	Daylight	H144	Inverted-variate	0.81015	0.78388	0.83433
Sanyo	Daylight	H144	Joint-patch	0.87534	0.80142	0.76574
Sanyo	Daylight	H144	Depthwise TCN	0.95405	0.87339	0.86283
Hanwha	Full	H12	Discrete recurrent	0.12387	0.12836	0.16202
Hanwha	Full	H12	Inverted-variate	0.13972	0.15415	0.15395
Hanwha	Full	H12	Joint-patch	0.19486	0.24511	0.17962
Hanwha	Full	H12	Depthwise TCN	0.12813	0.11327	0.12656
Hanwha	Daylight	H12	Discrete recurrent	0.20778	0.22419	0.28126
Hanwha	Daylight	H12	Inverted-variate	0.19702	0.22770	0.22425

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Daylight	H12	Joint-patch	0.25158	0.24992	0.22135
Hanwha	Daylight	H12	Depthwise TCN	0.18270	0.15758	0.18161
Hanwha	Full	H48	Discrete recurrent	0.27710	0.23620	0.32083
Hanwha	Full	H48	Inverted-variate	0.28438	0.32693	0.33336
Hanwha	Full	H48	Joint-patch	0.36652	0.34526	0.33407
Hanwha	Full	H48	Depthwise TCN	0.33142	0.33673	0.34221
Hanwha	Daylight	H48	Discrete recurrent	0.52195	0.46421	0.62989
Hanwha	Daylight	H48	Inverted-variate	0.47541	0.54087	0.52920
Hanwha	Daylight	H48	Joint-patch	0.59312	0.52110	0.52779
Hanwha	Daylight	H48	Depthwise TCN	0.52569	0.53522	0.53927
Hanwha	Full	H96	Discrete recurrent	0.40774	0.40206	0.45383
Hanwha	Full	H96	Inverted-variate	0.49253	0.52614	0.55511
Hanwha	Full	H96	Joint-patch	0.52264	0.46726	0.46749
Hanwha	Full	H96	Depthwise TCN	0.47434	0.48192	0.49329
Hanwha	Daylight	H96	Discrete recurrent	0.81616	0.83369	0.93289
Hanwha	Daylight	H96	Inverted-variate	0.87000	0.91131	0.93783
Hanwha	Daylight	H96	Joint-patch	0.92079	0.77157	0.79984
Hanwha	Daylight	H96	Depthwise TCN	0.81254	0.81988	0.83932
Hanwha	Full	H144	Discrete recurrent	0.40071	0.38910	0.42640
Hanwha	Full	H144	Inverted-variate	0.48388	0.52199	0.53621
Hanwha	Full	H144	Joint-patch	0.47310	0.44847	0.44631
Hanwha	Full	H144	Depthwise TCN	0.44980	0.47854	0.47476
Hanwha	Daylight	H144	Discrete recurrent	0.81786	0.82031	0.89037
Hanwha	Daylight	H144	Inverted-variate	0.84028	0.88713	0.89231
Hanwha	Daylight	H144	Joint-patch	0.82794	0.73477	0.76446
Hanwha	Daylight	H144	Depthwise TCN	0.74890	0.81601	0.79182
Qcells	Full	H12	Discrete recurrent	0.26715	0.44107	0.92759
Qcells	Full	H12	Inverted-variate	0.17351	0.23238	0.20505
Qcells	Full	H12	Joint-patch	0.38756	0.38343	0.33033
Qcells	Full	H12	Depthwise TCN	0.33873	0.18634	0.23193
Qcells	Daylight	H12	Discrete recurrent	0.51963	0.87320	1.85785
Qcells	Daylight	H12	Inverted-variate	0.28354	0.34031	0.33801
Qcells	Daylight	H12	Joint-patch	0.59609	0.67195	0.56727
Qcells	Daylight	H12	Depthwise TCN	0.62824	0.31843	0.44563
Qcells	Full	H48	Discrete recurrent	0.38057	0.54036	1.10494
Qcells	Full	H48	Inverted-variate	0.42459	0.46561	0.44915
Qcells	Full	H48	Joint-patch	0.60764	0.63207	0.60534
Qcells	Full	H48	Depthwise TCN	0.70893	0.59737	0.57099
Qcells	Daylight	H48	Discrete recurrent	0.75199	1.05716	2.25583
Qcells	Daylight	H48	Inverted-variate	0.77200	0.81193	0.79177
Qcells	Daylight	H48	Joint-patch	1.03945	1.18366	1.13374
Qcells	Daylight	H48	Depthwise TCN	1.38908	1.15015	1.12203
Qcells	Full	H96	Discrete recurrent	0.54199	0.65278	0.90353
Qcells	Full	H96	Inverted-variate	0.66567	0.68290	0.66126
Qcells	Full	H96	Joint-patch	0.74340	0.75035	0.71193
Qcells	Full	H96	Depthwise TCN	0.78189	0.82302	0.73178
Qcells	Daylight	H96	Discrete recurrent	1.11654	1.31308	1.84620
Qcells	Daylight	H96	Inverted-variate	1.24135	1.24596	1.19846
Qcells	Daylight	H96	Joint-patch	1.33358	1.41995	1.33442

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H96	Depthwise TCN	1.52256	1.61576	1.41140
Qcells	Full	H144	Discrete recurrent	0.49640	0.51181	0.48935
Qcells	Full	H144	Inverted-variate	0.52191	0.56507	0.51260
Qcells	Full	H144	Joint-patch	0.56861	0.53326	0.50553
Qcells	Full	H144	Depthwise TCN	0.49955	0.47670	0.47809
Qcells	Daylight	H144	Discrete recurrent	1.06417	1.04785	0.97894
Qcells	Daylight	H144	Inverted-variate	0.94392	0.98971	0.90037
Qcells	Daylight	H144	Joint-patch	0.97849	0.97913	0.92040
Qcells	Daylight	H144	Depthwise TCN	0.93775	0.88304	0.86744

Table S4: Primary per-seed R2 values.

Alt text: Per-seed coefficients of determination show horizon- and scope-dependent explained variance, including negative values where forecasts underperform the target mean. Values are descriptive because adjacent forecast origins overlap.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.99012	0.98966	0.99051
Sanyo	Full	H12	Inverted-variate	0.98379	0.98246	0.98810
Sanyo	Full	H12	Joint-patch	0.98017	0.98580	0.98890
Sanyo	Full	H12	Depthwise TCN	0.98743	0.99290	0.98508
Sanyo	Daylight	H12	Discrete recurrent	0.96350	0.96081	0.96469
Sanyo	Daylight	H12	Inverted-variate	0.95791	0.95588	0.96020
Sanyo	Daylight	H12	Joint-patch	0.94043	0.95655	0.96809
Sanyo	Daylight	H12	Depthwise TCN	0.95965	0.97592	0.95721
Sanyo	Full	H48	Discrete recurrent	0.90861	0.90941	0.90303
Sanyo	Full	H48	Inverted-variate	0.89887	0.91521	0.91574
Sanyo	Full	H48	Joint-patch	0.90236	0.92567	0.92301
Sanyo	Full	H48	Depthwise TCN	0.89986	0.90917	0.91200
Sanyo	Daylight	H48	Discrete recurrent	0.59798	0.60072	0.57216
Sanyo	Daylight	H48	Inverted-variate	0.59742	0.70551	0.67183
Sanyo	Daylight	H48	Joint-patch	0.59573	0.71394	0.68769
Sanyo	Daylight	H48	Depthwise TCN	0.60426	0.63806	0.66179
Sanyo	Full	H96	Discrete recurrent	0.81383	0.81722	0.78906
Sanyo	Full	H96	Inverted-variate	0.83419	0.85241	0.85188
Sanyo	Full	H96	Joint-patch	0.79010	0.83714	0.85173
Sanyo	Full	H96	Depthwise TCN	0.79050	0.83409	0.82312
Sanyo	Daylight	H96	Discrete recurrent	0.22692	0.24393	0.12087
Sanyo	Daylight	H96	Inverted-variate	0.39505	0.49534	0.46819
Sanyo	Daylight	H96	Joint-patch	0.16703	0.38723	0.43986
Sanyo	Daylight	H96	Depthwise TCN	0.20727	0.38910	0.34507
Sanyo	Full	H144	Discrete recurrent	0.81869	0.83965	0.80974
Sanyo	Full	H144	Inverted-variate	0.85662	0.86284	0.85625
Sanyo	Full	H144	Joint-patch	0.83160	0.85618	0.86775
Sanyo	Full	H144	Depthwise TCN	0.82702	0.85094	0.85096
Sanyo	Daylight	H144	Discrete recurrent	0.30739	0.39245	0.27621
Sanyo	Daylight	H144	Inverted-variate	0.53928	0.57615	0.53471
Sanyo	Daylight	H144	Joint-patch	0.39957	0.50855	0.54701
Sanyo	Daylight	H144	Depthwise TCN	0.41323	0.50541	0.50270

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Full	H12	Discrete recurrent	0.98994	0.98809	0.98471
Hanwha	Full	H12	Inverted-variate	0.98989	0.98780	0.98778
Hanwha	Full	H12	Joint-patch	0.98303	0.97656	0.98538
Hanwha	Full	H12	Depthwise TCN	0.99128	0.99240	0.99127
Hanwha	Daylight	H12	Discrete recurrent	0.96479	0.95696	0.94580
Hanwha	Daylight	H12	Inverted-variate	0.96898	0.96178	0.96210
Hanwha	Daylight	H12	Joint-patch	0.95397	0.95383	0.96269
Hanwha	Daylight	H12	Depthwise TCN	0.97447	0.97724	0.97347
Hanwha	Full	H48	Discrete recurrent	0.91295	0.91474	0.89469
Hanwha	Full	H48	Inverted-variate	0.92245	0.91005	0.91099
Hanwha	Full	H48	Joint-patch	0.90099	0.91483	0.91708
Hanwha	Full	H48	Depthwise TCN	0.91358	0.91121	0.90760
Hanwha	Daylight	H48	Discrete recurrent	0.65097	0.65314	0.57294
Hanwha	Daylight	H48	Inverted-variate	0.70142	0.65611	0.66810
Hanwha	Daylight	H48	Joint-patch	0.62249	0.68809	0.69251
Hanwha	Daylight	H48	Depthwise TCN	0.67831	0.67232	0.65654
Hanwha	Full	H96	Discrete recurrent	0.82583	0.80303	0.78172
Hanwha	Full	H96	Inverted-variate	0.81133	0.80494	0.78898
Hanwha	Full	H96	Joint-patch	0.79410	0.84324	0.83711
Hanwha	Full	H96	Depthwise TCN	0.83056	0.82993	0.82255
Hanwha	Daylight	H96	Discrete recurrent	0.33672	0.24166	0.15955
Hanwha	Daylight	H96	Inverted-variate	0.31390	0.30246	0.25794
Hanwha	Daylight	H96	Joint-patch	0.24331	0.44578	0.42006
Hanwha	Daylight	H96	Depthwise TCN	0.40080	0.41041	0.37877
Hanwha	Full	H144	Discrete recurrent	0.84051	0.83045	0.81835
Hanwha	Full	H144	Inverted-variate	0.82978	0.82222	0.81141
Hanwha	Full	H144	Joint-patch	0.82575	0.86041	0.85248
Hanwha	Full	H144	Depthwise TCN	0.85489	0.84483	0.84345
Hanwha	Daylight	H144	Discrete recurrent	0.43177	0.39155	0.34756
Hanwha	Daylight	H144	Inverted-variate	0.43445	0.42068	0.39739
Hanwha	Daylight	H144	Joint-patch	0.41750	0.54985	0.51736
Hanwha	Daylight	H144	Depthwise TCN	0.54036	0.50451	0.50586
Qcells	Full	H12	Discrete recurrent	0.94988	0.86178	0.35216
Qcells	Full	H12	Inverted-variate	0.97671	0.96878	0.97199
Qcells	Full	H12	Joint-patch	0.92817	0.91549	0.93806
Qcells	Full	H12	Depthwise TCN	0.92923	0.97608	0.95563
Qcells	Daylight	H12	Discrete recurrent	0.81396	0.48674	-1.41663
Qcells	Daylight	H12	Inverted-variate	0.91752	0.89828	0.90184
Qcells	Daylight	H12	Joint-patch	0.76075	0.69767	0.78069
Qcells	Daylight	H12	Depthwise TCN	0.74107	0.91487	0.83603
Qcells	Full	H48	Discrete recurrent	0.85828	0.78869	0.20558
Qcells	Full	H48	Inverted-variate	0.85194	0.84060	0.84465
Qcells	Full	H48	Joint-patch	0.79198	0.74781	0.76963
Qcells	Full	H48	Depthwise TCN	0.65567	0.74045	0.74340
Qcells	Daylight	H48	Discrete recurrent	0.37554	0.07240	-2.51653
Qcells	Daylight	H48	Inverted-variate	0.36053	0.31768	0.33533
Qcells	Daylight	H48	Joint-patch	0.11900	-0.09771	-0.00327
Qcells	Daylight	H48	Depthwise TCN	-0.51269	-0.13879	-0.12748
Qcells	Full	H96	Discrete recurrent	0.72851	0.66714	0.39676

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H96	Inverted-variate	0.67189	0.67404	0.69231
Qcells	Full	H96	Joint-patch	0.65344	0.62281	0.66015
Qcells	Full	H96	Depthwise TCN	0.54001	0.48761	0.59396
Qcells	Daylight	H96	Discrete recurrent	-0.08402	-0.31618	-1.40133
Qcells	Daylight	H96	Inverted-variate	-0.27312	-0.25659	-0.17520
Qcells	Daylight	H96	Joint-patch	-0.33529	-0.47135	-0.32141
Qcells	Daylight	H96	Depthwise TCN	-0.81087	-1.02099	-0.59047
Qcells	Full	H144	Discrete recurrent	0.78078	0.77078	0.78948
Qcells	Full	H144	Inverted-variate	0.80940	0.79931	0.82296
Qcells	Full	H144	Joint-patch	0.79847	0.80546	0.81892
Qcells	Full	H144	Depthwise TCN	0.82984	0.83272	0.84063
Qcells	Daylight	H144	Discrete recurrent	0.22854	0.20807	0.27726
Qcells	Daylight	H144	Inverted-variate	0.37117	0.34867	0.43114
Qcells	Daylight	H144	Joint-patch	0.34174	0.35260	0.39928
Qcells	Daylight	H144	Depthwise TCN	0.43312	0.44472	0.47782

Table S5: Primary per-seed Bias values.

Alt text: Per-seed mean errors identify underprediction and overprediction patterns across the three arrays, two scopes, and four forecast horizons. Positive values indicate overprediction, and negative values indicate underprediction.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	-0.01485	0.02522	-0.01926
Sanyo	Full	H12	Inverted-variate	0.06001	0.08298	-0.00450
Sanyo	Full	H12	Joint-patch	0.05334	0.07973	0.05282
Sanyo	Full	H12	Depthwise TCN	-0.06692	0.03892	0.16604
Sanyo	Daylight	H12	Discrete recurrent	-0.09490	0.04305	-0.06442
Sanyo	Daylight	H12	Inverted-variate	-0.06615	0.02881	-0.02995
Sanyo	Daylight	H12	Joint-patch	0.08527	0.10689	0.06451
Sanyo	Daylight	H12	Depthwise TCN	-0.22977	0.06247	0.18508
Sanyo	Full	H48	Discrete recurrent	-0.19674	-0.14171	-0.18156
Sanyo	Full	H48	Inverted-variate	-0.15590	0.00905	-0.13983
Sanyo	Full	H48	Joint-patch	-0.19055	-0.04334	-0.12676
Sanyo	Full	H48	Depthwise TCN	-0.19644	-0.14263	-0.07699
Sanyo	Daylight	H48	Discrete recurrent	-0.49498	-0.33281	-0.45605
Sanyo	Daylight	H48	Inverted-variate	-0.54239	-0.31128	-0.46493
Sanyo	Daylight	H48	Joint-patch	-0.44354	-0.25136	-0.35105
Sanyo	Daylight	H48	Depthwise TCN	-0.58612	-0.43602	-0.37384
Sanyo	Full	H96	Discrete recurrent	-0.33184	-0.30250	-0.35482
Sanyo	Full	H96	Inverted-variate	-0.21373	-0.09706	-0.20703
Sanyo	Full	H96	Joint-patch	-0.36865	-0.22148	-0.22546
Sanyo	Full	H96	Depthwise TCN	-0.35401	-0.19358	-0.20143
Sanyo	Daylight	H96	Discrete recurrent	-0.80950	-0.71352	-0.84974
Sanyo	Daylight	H96	Inverted-variate	-0.66186	-0.54659	-0.61488
Sanyo	Daylight	H96	Joint-patch	-0.84300	-0.59778	-0.54960
Sanyo	Daylight	H96	Depthwise TCN	-0.90937	-0.60478	-0.67695
Sanyo	Full	H144	Discrete recurrent	-0.30150	-0.24728	-0.28443
Sanyo	Full	H144	Inverted-variate	-0.12415	-0.07899	-0.19460

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H144	Joint-patch	-0.22895	-0.16462	-0.18296
Sanyo	Full	H144	Depthwise TCN	-0.25089	-0.17817	-0.16368
Sanyo	Daylight	H144	Discrete recurrent	-0.76490	-0.61458	-0.72355
Sanyo	Daylight	H144	Inverted-variate	-0.45553	-0.45708	-0.55347
Sanyo	Daylight	H144	Joint-patch	-0.57689	-0.49772	-0.44642
Sanyo	Daylight	H144	Depthwise TCN	-0.70901	-0.59856	-0.57486
Hanwha	Full	H12	Discrete recurrent	-0.07223	-0.06327	-0.12246
Hanwha	Full	H12	Inverted-variate	-0.00845	-0.01508	-0.01574
Hanwha	Full	H12	Joint-patch	0.09252	-0.19886	-0.05599
Hanwha	Full	H12	Depthwise TCN	-0.02420	0.01298	0.03659
Hanwha	Daylight	H12	Discrete recurrent	-0.13923	-0.12002	-0.21837
Hanwha	Daylight	H12	Inverted-variate	0.00506	-0.08452	-0.07401
Hanwha	Daylight	H12	Joint-patch	0.06763	-0.15859	0.00038
Hanwha	Daylight	H12	Depthwise TCN	-0.03043	-0.00929	0.05944
Hanwha	Full	H48	Discrete recurrent	-0.21430	-0.17656	-0.27482
Hanwha	Full	H48	Inverted-variate	-0.13472	-0.18278	-0.14005
Hanwha	Full	H48	Joint-patch	-0.15171	-0.23340	-0.19411
Hanwha	Full	H48	Depthwise TCN	-0.19526	-0.16190	-0.18508
Hanwha	Daylight	H48	Discrete recurrent	-0.46980	-0.38290	-0.56927
Hanwha	Daylight	H48	Inverted-variate	-0.35116	-0.44743	-0.40447
Hanwha	Daylight	H48	Joint-patch	-0.45830	-0.40349	-0.37726
Hanwha	Daylight	H48	Depthwise TCN	-0.41941	-0.45174	-0.41057
Hanwha	Full	H96	Discrete recurrent	-0.32869	-0.33349	-0.39879
Hanwha	Full	H96	Inverted-variate	-0.32324	-0.34001	-0.34498
Hanwha	Full	H96	Joint-patch	-0.33245	-0.29819	-0.28419
Hanwha	Full	H96	Depthwise TCN	-0.26653	-0.22095	-0.26397
Hanwha	Daylight	H96	Discrete recurrent	-0.72991	-0.74335	-0.86099
Hanwha	Daylight	H96	Inverted-variate	-0.72901	-0.76600	-0.78191
Hanwha	Daylight	H96	Joint-patch	-0.77479	-0.55500	-0.59110
Hanwha	Daylight	H96	Depthwise TCN	-0.62178	-0.63182	-0.64313
Hanwha	Full	H144	Discrete recurrent	-0.27785	-0.26207	-0.31324
Hanwha	Full	H144	Inverted-variate	-0.27110	-0.30315	-0.28482
Hanwha	Full	H144	Joint-patch	-0.20469	-0.24152	-0.21510
Hanwha	Full	H144	Depthwise TCN	-0.17934	-0.19233	-0.18866
Hanwha	Daylight	H144	Discrete recurrent	-0.62704	-0.60357	-0.69016
Hanwha	Daylight	H144	Inverted-variate	-0.60275	-0.65849	-0.64942
Hanwha	Daylight	H144	Joint-patch	-0.52855	-0.43327	-0.48145
Hanwha	Daylight	H144	Depthwise TCN	-0.46343	-0.56530	-0.50861
Qcells	Full	H12	Discrete recurrent	-0.13648	-0.30177	-0.82254
Qcells	Full	H12	Inverted-variate	-0.01945	-0.08940	-0.05842
Qcells	Full	H12	Joint-patch	-0.04125	-0.14744	-0.20265
Qcells	Full	H12	Depthwise TCN	-0.19140	-0.08991	0.01064
Qcells	Daylight	H12	Discrete recurrent	-0.30502	-0.68714	-1.67725
Qcells	Daylight	H12	Inverted-variate	-0.01510	-0.10341	-0.09454
Qcells	Daylight	H12	Joint-patch	-0.30270	-0.42471	-0.34917
Qcells	Daylight	H12	Depthwise TCN	-0.46033	-0.15339	0.02249
Qcells	Full	H48	Discrete recurrent	-0.26981	-0.42263	-1.02588
Qcells	Full	H48	Inverted-variate	-0.28195	-0.33935	-0.29151
Qcells	Full	H48	Joint-patch	-0.32374	-0.46845	-0.52214

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H48	Depthwise TCN	-0.59271	-0.51875	-0.42368
Qcells	Daylight	H48	Discrete recurrent	-0.60787	-0.97543	-2.18057
Qcells	Daylight	H48	Inverted-variate	-0.59994	-0.65309	-0.61191
Qcells	Daylight	H48	Joint-patch	-0.92710	-1.09891	-1.06507
Qcells	Daylight	H48	Depthwise TCN	-1.33813	-1.10355	-0.93068
Qcells	Full	H96	Discrete recurrent	-0.41366	-0.49742	-0.76850
Qcells	Full	H96	Inverted-variate	-0.44038	-0.45492	-0.37560
Qcells	Full	H96	Joint-patch	-0.45584	-0.54116	-0.55976
Qcells	Full	H96	Depthwise TCN	-0.59562	-0.65654	-0.54134
Qcells	Daylight	H96	Discrete recurrent	-0.95209	-1.17593	-1.69645
Qcells	Daylight	H96	Inverted-variate	-0.96993	-0.98873	-0.91253
Qcells	Daylight	H96	Joint-patch	-1.15251	-1.26438	-1.17279
Qcells	Daylight	H96	Depthwise TCN	-1.38535	-1.48752	-1.19143
Qcells	Full	H144	Discrete recurrent	-0.35576	-0.27276	-0.26605
Qcells	Full	H144	Inverted-variate	-0.28983	-0.33920	-0.20222
Qcells	Full	H144	Joint-patch	-0.25484	-0.28206	-0.29352
Qcells	Full	H144	Depthwise TCN	-0.25781	-0.22607	-0.25775
Qcells	Daylight	H144	Discrete recurrent	-0.85934	-0.71506	-0.62348
Qcells	Daylight	H144	Inverted-variate	-0.63709	-0.71159	-0.54233
Qcells	Daylight	H144	Joint-patch	-0.69371	-0.70131	-0.62555
Qcells	Daylight	H144	Depthwise TCN	-0.68415	-0.55354	-0.55718

Table S6: Primary per-seed range_nRMSE values.

Alt text: Per-seed Train-range normalized RMSE values permit within-array comparison of error growth while avoiding unsupported AC-capacity normalization. The denominator is fitted only on Train and differs by array.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.03417	0.03495	0.03348
Sanyo	Full	H12	Inverted-variate	0.04377	0.04553	0.03750
Sanyo	Full	H12	Joint-patch	0.04841	0.04097	0.03622
Sanyo	Full	H12	Depthwise TCN	0.03855	0.02897	0.04199
Sanyo	Daylight	H12	Discrete recurrent	0.04886	0.05064	0.04806
Sanyo	Daylight	H12	Inverted-variate	0.05248	0.05373	0.05103
Sanyo	Daylight	H12	Joint-patch	0.06242	0.05331	0.04569
Sanyo	Daylight	H12	Depthwise TCN	0.05138	0.03969	0.05291
Sanyo	Full	H48	Discrete recurrent	0.10678	0.10631	0.10999
Sanyo	Full	H48	Inverted-variate	0.11232	0.10285	0.10253
Sanyo	Full	H48	Joint-patch	0.11037	0.09630	0.09800
Sanyo	Full	H48	Depthwise TCN	0.11177	0.10645	0.10477
Sanyo	Daylight	H48	Discrete recurrent	0.15678	0.15624	0.16174
Sanyo	Daylight	H48	Inverted-variate	0.15689	0.13418	0.14165
Sanyo	Daylight	H48	Joint-patch	0.15722	0.13225	0.13818
Sanyo	Daylight	H48	Depthwise TCN	0.15555	0.14876	0.14380
Sanyo	Full	H96	Discrete recurrent	0.14995	0.14859	0.15962
Sanyo	Full	H96	Inverted-variate	0.14152	0.13352	0.13375
Sanyo	Full	H96	Joint-patch	0.15922	0.14025	0.13383
Sanyo	Full	H96	Depthwise TCN	0.15907	0.14156	0.14616
Sanyo	Daylight	H96	Discrete recurrent	0.22262	0.22016	0.23740

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Daylight	H96	Inverted-variate	0.19693	0.17987	0.18464
Sanyo	Daylight	H96	Joint-patch	0.23108	0.19820	0.18950
Sanyo	Daylight	H96	Depthwise TCN	0.22543	0.19790	0.20490
Sanyo	Full	H144	Discrete recurrent	0.14497	0.13634	0.14850
Sanyo	Full	H144	Inverted-variate	0.12892	0.12609	0.12908
Sanyo	Full	H144	Joint-patch	0.13971	0.12911	0.12381
Sanyo	Full	H144	Depthwise TCN	0.14160	0.13145	0.13144
Sanyo	Daylight	H144	Discrete recurrent	0.21658	0.20284	0.22140
Sanyo	Daylight	H144	Inverted-variate	0.17664	0.16942	0.17751
Sanyo	Daylight	H144	Joint-patch	0.20165	0.18243	0.17515
Sanyo	Daylight	H144	Depthwise TCN	0.19934	0.18301	0.18352
Hanwha	Full	H12	Discrete recurrent	0.03308	0.03600	0.04078
Hanwha	Full	H12	Inverted-variate	0.03316	0.03643	0.03646
Hanwha	Full	H12	Joint-patch	0.04297	0.05050	0.03988
Hanwha	Full	H12	Depthwise TCN	0.03079	0.02876	0.03081
Hanwha	Daylight	H12	Discrete recurrent	0.04703	0.05200	0.05835
Hanwha	Daylight	H12	Inverted-variate	0.04414	0.04900	0.04879
Hanwha	Daylight	H12	Joint-patch	0.05377	0.05385	0.04841
Hanwha	Daylight	H12	Depthwise TCN	0.04005	0.03781	0.04082
Hanwha	Full	H48	Discrete recurrent	0.09950	0.09847	0.10944
Hanwha	Full	H48	Inverted-variate	0.09391	0.10114	0.10061
Hanwha	Full	H48	Joint-patch	0.10611	0.09842	0.09711
Hanwha	Full	H48	Depthwise TCN	0.09914	0.10049	0.10251
Hanwha	Daylight	H48	Discrete recurrent	0.14534	0.14489	0.16077
Hanwha	Daylight	H48	Inverted-variate	0.13443	0.14427	0.14173
Hanwha	Daylight	H48	Joint-patch	0.15115	0.13740	0.13642
Hanwha	Daylight	H48	Depthwise TCN	0.13953	0.14083	0.14418
Hanwha	Full	H96	Discrete recurrent	0.13866	0.14746	0.15523
Hanwha	Full	H96	Inverted-variate	0.14432	0.14674	0.15263
Hanwha	Full	H96	Joint-patch	0.15076	0.13155	0.13410
Hanwha	Full	H96	Depthwise TCN	0.13677	0.13702	0.13996
Hanwha	Daylight	H96	Discrete recurrent	0.20444	0.21860	0.23013
Hanwha	Daylight	H96	Inverted-variate	0.20792	0.20965	0.21624
Hanwha	Daylight	H96	Joint-patch	0.21836	0.18687	0.19116
Hanwha	Daylight	H96	Depthwise TCN	0.19431	0.19275	0.19785
Hanwha	Full	H144	Discrete recurrent	0.13050	0.13455	0.13927
Hanwha	Full	H144	Inverted-variate	0.13482	0.13778	0.14191
Hanwha	Full	H144	Joint-patch	0.13640	0.12208	0.12550
Hanwha	Full	H144	Depthwise TCN	0.12448	0.12872	0.12929
Hanwha	Daylight	H144	Discrete recurrent	0.19335	0.20007	0.20718
Hanwha	Daylight	H144	Inverted-variate	0.19289	0.19523	0.19911
Hanwha	Daylight	H144	Joint-patch	0.19576	0.17209	0.17819
Hanwha	Daylight	H144	Depthwise TCN	0.17390	0.18055	0.18030
Qcells	Full	H12	Discrete recurrent	0.07332	0.12176	0.26361
Qcells	Full	H12	Inverted-variate	0.04999	0.05787	0.05482
Qcells	Full	H12	Joint-patch	0.08778	0.09521	0.08151
Qcells	Full	H12	Depthwise TCN	0.08713	0.05065	0.06899
Qcells	Daylight	H12	Discrete recurrent	0.10642	0.17677	0.38357
Qcells	Daylight	H12	Inverted-variate	0.07086	0.07869	0.07731

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H12	Joint-patch	0.12069	0.13567	0.11555
Qcells	Daylight	H12	Depthwise TCN	0.12556	0.07199	0.09991
Qcells	Full	H48	Discrete recurrent	0.12822	0.15657	0.30358
Qcells	Full	H48	Inverted-variate	0.13106	0.13598	0.13425
Qcells	Full	H48	Joint-patch	0.15535	0.17104	0.16348
Qcells	Full	H48	Depthwise TCN	0.19986	0.17352	0.17253
Qcells	Daylight	H48	Discrete recurrent	0.18663	0.22746	0.44287
Qcells	Daylight	H48	Inverted-variate	0.18886	0.19508	0.19254
Qcells	Daylight	H48	Joint-patch	0.22167	0.24744	0.23655
Qcells	Daylight	H48	Depthwise TCN	0.29047	0.25202	0.25077
Qcells	Full	H96	Discrete recurrent	0.17328	0.19187	0.25830
Qcells	Full	H96	Inverted-variate	0.19050	0.18988	0.18448
Qcells	Full	H96	Joint-patch	0.19578	0.20425	0.19388
Qcells	Full	H96	Depthwise TCN	0.22556	0.23806	0.21192
Qcells	Daylight	H96	Discrete recurrent	0.25569	0.28174	0.38055
Qcells	Daylight	H96	Inverted-variate	0.27709	0.27529	0.26622
Qcells	Daylight	H96	Joint-patch	0.28378	0.29788	0.28230
Qcells	Daylight	H96	Depthwise TCN	0.33047	0.34912	0.30971
Qcells	Full	H144	Discrete recurrent	0.15035	0.15374	0.14734
Qcells	Full	H144	Inverted-variate	0.14019	0.14386	0.13512
Qcells	Full	H144	Joint-patch	0.14416	0.14164	0.13665
Qcells	Full	H144	Depthwise TCN	0.13246	0.13134	0.12819
Qcells	Daylight	H144	Discrete recurrent	0.22649	0.22948	0.21922
Qcells	Daylight	H144	Inverted-variate	0.20448	0.20811	0.19449
Qcells	Daylight	H144	Joint-patch	0.20921	0.20748	0.19986
Qcells	Daylight	H144	Depthwise TCN	0.19415	0.19215	0.18634

Table S7: Primary per-seed RMSE_skill values.

Alt text: Per-seed RMSE skill relative to Last-value Persistence quantifies improvement beyond local continuity; positive values favor the neural implementation. Every value uses the same origins, labels, and masks as its reference.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.55381	0.54361	0.56280
Sanyo	Full	H12	Inverted-variate	0.42848	0.40540	0.51032
Sanyo	Full	H12	Joint-patch	0.36784	0.46504	0.52699
Sanyo	Full	H12	Depthwise TCN	0.49666	0.62169	0.45164
Sanyo	Daylight	H12	Discrete recurrent	0.55629	0.54019	0.56358
Sanyo	Daylight	H12	Inverted-variate	0.52348	0.51213	0.53664
Sanyo	Daylight	H12	Joint-patch	0.43314	0.51588	0.58512
Sanyo	Daylight	H12	Depthwise TCN	0.53347	0.63962	0.51956
Sanyo	Full	H48	Discrete recurrent	0.55058	0.55255	0.53706
Sanyo	Full	H48	Inverted-variate	0.52724	0.56711	0.56847
Sanyo	Full	H48	Joint-patch	0.53546	0.59469	0.58751
Sanyo	Full	H48	Depthwise TCN	0.52955	0.55195	0.55900
Sanyo	Daylight	H48	Discrete recurrent	0.53089	0.53249	0.51606
Sanyo	Daylight	H48	Inverted-variate	0.53056	0.59850	0.57616
Sanyo	Daylight	H48	Joint-patch	0.52958	0.60429	0.58653

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Daylight	H48	Depthwise TCN	0.53457	0.55489	0.56973
Sanyo	Full	H96	Discrete recurrent	0.60553	0.60913	0.58010
Sanyo	Full	H96	Inverted-variate	0.62772	0.64877	0.64814
Sanyo	Full	H96	Joint-patch	0.58114	0.63104	0.64795
Sanyo	Full	H96	Depthwise TCN	0.58154	0.62760	0.61550
Sanyo	Daylight	H96	Discrete recurrent	0.57233	0.57706	0.54394
Sanyo	Daylight	H96	Inverted-variate	0.62169	0.65447	0.64529
Sanyo	Daylight	H96	Joint-patch	0.55608	0.61925	0.63597
Sanyo	Daylight	H96	Depthwise TCN	0.56693	0.61983	0.60637
Sanyo	Full	H144	Discrete recurrent	0.67824	0.69741	0.67040
Sanyo	Full	H144	Inverted-variate	0.71387	0.72014	0.71350
Sanyo	Full	H144	Joint-patch	0.68991	0.71343	0.72520
Sanyo	Full	H144	Depthwise TCN	0.68572	0.70826	0.70828
Sanyo	Daylight	H144	Discrete recurrent	0.64239	0.66507	0.63443
Sanyo	Daylight	H144	Inverted-variate	0.70834	0.72025	0.70689
Sanyo	Daylight	H144	Joint-patch	0.66704	0.69876	0.71079
Sanyo	Daylight	H144	Depthwise TCN	0.67084	0.69781	0.69698
Hanwha	Full	H12	Discrete recurrent	0.55574	0.51655	0.45229
Hanwha	Full	H12	Inverted-variate	0.55474	0.51080	0.51031
Hanwha	Full	H12	Joint-patch	0.42297	0.32186	0.46437
Hanwha	Full	H12	Depthwise TCN	0.58645	0.61382	0.58622
Hanwha	Daylight	H12	Discrete recurrent	0.56394	0.51785	0.45895
Hanwha	Daylight	H12	Inverted-variate	0.59070	0.54567	0.54755
Hanwha	Daylight	H12	Joint-patch	0.50142	0.50064	0.55112
Hanwha	Daylight	H12	Depthwise TCN	0.62865	0.64938	0.62148
Hanwha	Full	H48	Discrete recurrent	0.57112	0.57557	0.52829
Hanwha	Full	H48	Inverted-variate	0.59521	0.56405	0.56633
Hanwha	Full	H48	Joint-patch	0.54262	0.57578	0.58142
Hanwha	Full	H48	Depthwise TCN	0.57268	0.56686	0.55815
Hanwha	Daylight	H48	Discrete recurrent	0.54906	0.55046	0.50119
Hanwha	Daylight	H48	Inverted-variate	0.58292	0.55239	0.56026
Hanwha	Daylight	H48	Joint-patch	0.53102	0.57371	0.57674
Hanwha	Daylight	H48	Depthwise TCN	0.56708	0.56307	0.55267
Hanwha	Full	H96	Discrete recurrent	0.62830	0.60473	0.58390
Hanwha	Full	H96	Inverted-variate	0.61315	0.60664	0.59087
Hanwha	Full	H96	Joint-patch	0.59587	0.64737	0.64055
Hanwha	Full	H96	Depthwise TCN	0.63338	0.63270	0.62482
Hanwha	Daylight	H96	Discrete recurrent	0.58419	0.55539	0.53194
Hanwha	Daylight	H96	Inverted-variate	0.57710	0.57359	0.56019
Hanwha	Daylight	H96	Joint-patch	0.55587	0.61991	0.61119
Hanwha	Daylight	H96	Depthwise TCN	0.60478	0.60797	0.59758
Hanwha	Full	H144	Discrete recurrent	0.70299	0.69377	0.68303
Hanwha	Full	H144	Inverted-variate	0.69316	0.68642	0.67703
Hanwha	Full	H144	Joint-patch	0.68956	0.72214	0.71436
Hanwha	Full	H144	Depthwise TCN	0.71670	0.70704	0.70574
Hanwha	Daylight	H144	Discrete recurrent	0.65561	0.64363	0.63098
Hanwha	Daylight	H144	Inverted-variate	0.65643	0.65227	0.64535
Hanwha	Daylight	H144	Joint-patch	0.65132	0.69348	0.68261
Hanwha	Daylight	H144	Depthwise TCN	0.69026	0.67841	0.67885

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H12	Discrete recurrent	0.05980	-0.56142	-2.38038
Qcells	Full	H12	Inverted-variate	0.35900	0.25792	0.29709
Qcells	Full	H12	Joint-patch	-0.12558	-0.22089	-0.04527
Qcells	Full	H12	Depthwise TCN	-0.11728	0.35050	0.11532
Qcells	Daylight	H12	Discrete recurrent	0.05756	-0.56540	-2.39674
Qcells	Daylight	H12	Inverted-variate	0.37248	0.30313	0.31542
Qcells	Daylight	H12	Joint-patch	-0.06876	-0.20143	-0.02325
Qcells	Daylight	H12	Depthwise TCN	-0.11186	0.36248	0.11521
Qcells	Full	H48	Discrete recurrent	0.40800	0.27713	-0.40160
Qcells	Full	H48	Inverted-variate	0.39491	0.37217	0.38020
Qcells	Full	H48	Joint-patch	0.28278	0.21030	0.24524
Qcells	Full	H48	Depthwise TCN	0.07725	0.19886	0.20342
Qcells	Daylight	H48	Discrete recurrent	0.40872	0.27935	-0.40314
Qcells	Daylight	H48	Inverted-variate	0.40165	0.38193	0.38998
Qcells	Daylight	H48	Joint-patch	0.29769	0.21605	0.25054
Qcells	Daylight	H48	Depthwise TCN	0.07972	0.20152	0.20550
Qcells	Full	H96	Discrete recurrent	0.49082	0.43619	0.24100
Qcells	Full	H96	Inverted-variate	0.44023	0.44207	0.45793
Qcells	Full	H96	Joint-patch	0.42471	0.39982	0.43031
Qcells	Full	H96	Depthwise TCN	0.33721	0.30048	0.37729
Qcells	Daylight	H96	Discrete recurrent	0.49248	0.44077	0.24463
Qcells	Daylight	H96	Inverted-variate	0.44999	0.45357	0.47157
Qcells	Daylight	H96	Joint-patch	0.43672	0.40872	0.43966
Qcells	Daylight	H96	Depthwise TCN	0.34404	0.30703	0.38525
Qcells	Full	H144	Discrete recurrent	0.62459	0.61612	0.63211
Qcells	Full	H144	Inverted-variate	0.64995	0.64080	0.66263
Qcells	Full	H144	Joint-patch	0.64005	0.64635	0.65881
Qcells	Full	H144	Depthwise TCN	0.66925	0.67206	0.67992
Qcells	Daylight	H144	Discrete recurrent	0.62674	0.62182	0.63871
Qcells	Daylight	H144	Inverted-variate	0.66300	0.65703	0.67948
Qcells	Daylight	H144	Joint-patch	0.65521	0.65806	0.67062
Qcells	Daylight	H144	Depthwise TCN	0.68003	0.68333	0.69291

Table S8: Primary horizon-specific evaluation support.

Alt text: Forecast-origin and valid-target counts decrease with horizon and are shared by every method within each array-specific comparison. Full and daylight scopes report their target support separately.

Array	Scope	Horizon	Forecast origins	Valid target points
Sanyo	Full	H12	6,589	79,068
Sanyo	Daylight	H12	3,201	36,190
Sanyo	Full	H48	5,905	283,440
Sanyo	Daylight	H48	3,389	129,528
Sanyo	Full	H96	5,024	482,304
Sanyo	Daylight	H96	3,629	215,712
Sanyo	Full	H144	4,160	599,040
Sanyo	Daylight	H144	3,869	259,345
Hanwha	Full	H12	6,625	79,500
Hanwha	Daylight	H12	3,225	36,421

Array	Scope	Horizon	Forecast origins	Valid target points
Hanwha	Full	H48	6,049	290,352
Hanwha	Daylight	H48	3,522	132,649
Hanwha	Full	H96	5,281	506,976
Hanwha	Daylight	H96	3,881	228,420
Hanwha	Full	H144	4,521	651,024
Hanwha	Daylight	H144	4,225	287,368
Qcells	Full	H12	6,463	77,556
Qcells	Daylight	H12	3,173	36,504
Qcells	Full	H48	5,491	263,568
Qcells	Daylight	H48	3,071	123,447
Qcells	Full	H96	4,227	405,792
Qcells	Daylight	H96	2,927	184,959
Qcells	Full	H144	2,996	431,424
Qcells	Daylight	H144	2,800	187,950

Table S9: Complete-H144-origin sensitivity: Train-range nRMSE for every evaluated prefix. All horizons within a row use origins that possess a complete valid H144 target.

Alt text: Restricting every prefix to complete H144 origins preserves the main pattern of horizon-dependent error without changing the primary horizon-specific analysis. This analysis is secondary to the horizon-specific evaluation.

Array	Scope	Model	H12	H48	H96	H144
Sanyo	Full	Last-value	0.0491	0.1857	0.3539	0.4506
Sanyo	Full	Discrete recurrent	0.0230	0.0958	0.1458	0.1433
Sanyo	Full	Inverted-variate	0.0358	0.0948	0.1261	0.1280
Sanyo	Full	Joint-patch	0.0313	0.0907	0.1346	0.1309
Sanyo	Full	Depthwise TCN	0.0265	0.0950	0.1358	0.1348
Sanyo	Daylight	Last-value	0.1092	0.3385	0.5383	0.6056
Sanyo	Daylight	Discrete recurrent	0.0488	0.1929	0.2490	0.2136
Sanyo	Daylight	Inverted-variate	0.0525	0.1655	0.1921	0.1745
Sanyo	Daylight	Joint-patch	0.0492	0.1680	0.2175	0.1864
Sanyo	Daylight	Depthwise TCN	0.0405	0.1712	0.2137	0.1886
Hanwha	Full	Last-value	0.0591	0.2003	0.3546	0.4394
Hanwha	Full	Discrete recurrent	0.0309	0.0920	0.1381	0.1348
Hanwha	Full	Inverted-variate	0.0316	0.0907	0.1396	0.1382
Hanwha	Full	Joint-patch	0.0419	0.0937	0.1324	0.1280
Hanwha	Full	Depthwise TCN	0.0284	0.0936	0.1299	0.1275
Hanwha	Daylight	Last-value	0.1109	0.3252	0.4982	0.5614
Hanwha	Daylight	Discrete recurrent	0.0562	0.1631	0.2231	0.2002
Hanwha	Daylight	Inverted-variate	0.0509	0.1531	0.2153	0.1957
Hanwha	Daylight	Joint-patch	0.0539	0.1562	0.2052	0.1820
Hanwha	Daylight	Depthwise TCN	0.0453	0.1553	0.1974	0.1782
Qcells	Full	Last-value	0.0007	0.1037	0.2873	0.4005
Qcells	Full	Discrete recurrent	0.0138	0.0849	0.1527	0.1505
Qcells	Full	Inverted-variate	0.0207	0.0829	0.1411	0.1397
Qcells	Full	Joint-patch	0.0291	0.0849	0.1452	0.1408
Qcells	Full	Depthwise TCN	0.0129	0.0818	0.1334	0.1307

Array	Scope	Model	H12	H48	H96	H144
Qcells	Daylight	Last-value	0.0179	0.3209	0.5541	0.6068
Qcells	Daylight	Discrete recurrent	0.0219	0.2552	0.2898	0.2251
Qcells	Daylight	Inverted-variate	0.0315	0.2371	0.2574	0.2024
Qcells	Daylight	Joint-patch	0.0222	0.2382	0.2671	0.2055
Qcells	Daylight	Depthwise TCN	0.0201	0.2407	0.2456	0.1909

Table S10: Daily-matched RMSE (kW); the Daily-valid point mask is identical for all methods.

Alt text: On identical Daily-valid target points, Daily Persistence has lower RMSE than the post hoc neural envelope in 22 of 24 comparisons; the exceptions are Hanwha H12 full and daylight.

Array	Scope	Horizon	Daily	Last	Discrete	Inverted	Joint-patch	Depthwise
Sanyo	Full	H12	0.1752	0.4637	0.2072	0.2560	0.2535	0.2212
Sanyo	Daylight	H12	0.2589	0.6671	0.2980	0.3175	0.3260	0.2907
Sanyo	Full	H48	0.1595	1.4371	0.6527	0.6417	0.6153	0.6524
Sanyo	Daylight	H48	0.2359	2.0246	0.9587	0.8738	0.8636	0.9049
Sanyo	Full	H96	0.1291	2.3001	0.9256	0.8258	0.8753	0.9026
Sanyo	Daylight	H96	0.1929	3.1534	1.3735	1.1337	1.2495	1.2686
Sanyo	Full	H144	0.1181	2.7281	0.8677	0.7753	0.7926	0.8166
Sanyo	Daylight	H144	0.1794	3.6688	1.2940	1.0572	1.1293	1.1427
Hanwha	Full	H12	0.1852	0.4339	0.2136	0.2061	0.2591	0.1756
Hanwha	Daylight	H12	0.2734	0.6286	0.3058	0.2758	0.3032	0.2306
Hanwha	Full	H48	0.1932	1.3501	0.5977	0.5747	0.5864	0.5874
Hanwha	Daylight	H48	0.2856	1.8787	0.8763	0.8169	0.8257	0.8249
Hanwha	Full	H96	0.2035	2.1714	0.8582	0.8626	0.8096	0.8043
Hanwha	Daylight	H96	0.3029	2.8658	1.2691	1.2315	1.1588	1.1364
Hanwha	Full	H144	0.2031	2.5597	0.7860	0.8054	0.7460	0.7431
Hanwha	Daylight	H144	0.3053	3.2725	1.1669	1.1410	1.0609	1.0390
Qcells	Full	H12	0.2805	0.4708	0.9232	0.3274	0.5323	0.4161
Qcells	Daylight	H12	0.4089	0.6817	1.3417	0.4565	0.7484	0.5986
Qcells	Full	H48	0.2806	1.3077	1.1842	0.8076	0.9859	1.0987
Qcells	Daylight	H48	0.4099	1.9054	1.7244	1.1600	1.4200	1.5963
Qcells	Full	H96	0.2553	2.0549	1.2549	1.1368	1.1953	1.3596
Qcells	Daylight	H96	0.3780	3.0413	1.8472	1.6472	1.7385	1.9907
Qcells	Full	H144	0.2436	2.4185	0.9087	0.8436	0.8502	0.7889
Qcells	Daylight	H144	0.3690	3.6631	1.3587	1.2216	1.2407	1.1523

6 External data, time coordinates, and limitations

Yulara is one 106.6 kW combined facility, not three independent subarray sites. Its file spans 2016-04-01 23:45 to 2026-02-19 23:55. The 2017 segment has 105,122 distinct ordered rows: 105,120 regular five-minute timestamps and two off-grid records. The filename's 2016 label is not the experiment year. The off-grid timestamps are 2017-02-22 15:30:02 (75.150801518689 kW) and 2017-08-07 05:20:01 (-0.047513291616942 kW); temperature and GHI are missing in both. Both have existing floor and ceiling grid records. They are excluded without rounding, filling the regular record, or modifying the source.

Yulara retains its provider local coordinate, with status `PROVIDER_LOCAL_TIME_OFFSET_UNCONFIRMED`. An observed value at raw timestamp t is available at $t + 5$ min. This fixed coordinate has no introduced DST jumps. The official download and system pages identify the facility and power variable. Temperature follows the Celsius field name; GHI retains the provider-native numbers. Detailed aggregation, sensor uncertainty, and invalid-code definitions remain partial. This is a nonblocking documentation limitation under the frozen protocol; it does not authorize guessed unit conversions or absolute cross-site GHI comparisons.

NIST comprises 365 one-minute CSVs with consistent headers and 525,595 distinct ordered timestamps from 2017-01-01 00:00 to 2017-12-31 23:59. Five minutes are absent: 2017-10-20 23:59 and 2017-10-21 00:00, 00:01, 00:02, and 00:03, all at UTC−05:00. Parsing, bins, grids, splits, and Daily joins retain this timezone-aware fixed EST/LST coordinate. There are no DST missing or repeated hours.

At an end time T, each variable uses exactly five distinct finite minute observations in $[T - 5\text{min}, T)$. Bins have a midnight origin, five-minute anchor, left closure, right label, and availability T. An incomplete variable has a missing bin value, never a partial mean. The official CSV supplies `Pyra1_Wm2_Avg` directly, and no `Pyra1_mV_Avg` column is present. The AC meter field `PwrMtrP_kW_Avg` supplies input power and labels. `InvPAC_kW_Avg` has 4,712 entries equal to -999 and is excluded from both inputs and labels.

Train-only timing diagnostics compared energy increments with average power and inspected power/GHI correspondence. The unshifted power comparison had mean absolute energy discrepancy 0.164 kWh, versus 0.226 kWh at either adjacent-minute alignment. Power/GHI correlation was 0.972 without a five-minute displacement versus approximately 0.929 with either displacement. These are diagnostic summaries, not predictive results or a Test-based rule search. The conservative interval-end availability remains the frozen rule.

No complete NIST 2017 Ground operation log was obtained. This is recorded as a nonblocking documentation limitation, not evidence that no maintenance, snow, outage, or instrument event occurred. Finite negative power and GHI, and high-GHI/nonpositive-power observations, are retained. NIST has 14,418 negative power and 271,221 negative GHI minute records; 751 records have native GHI above 500 with nonpositive power. Yulara's 2017 raw segment has 52,807 negative power and 50,504 negative GHI records, with zero in the analogous high-GHI/nonpositive-power category. These are descriptive counts, not quality filters or cross-site irradiance comparisons.

Table S11: External raw-field quality inventory for the 2017 segment, before availability shifts and aggregation. Nonnum: non-numeric; Inf: either infinity. Minima, medians, 95th percentiles and maxima use finite values.

Site	Variable	Null	Nonnum	Inf	Min	Median	P95	Max
Yulara	AC power (kW)	64	0	0	-0.059	-0.035	88.133	109.903
Yulara	Temperature (°C)	2950	0	0	-2.400	22.433	36.100	43.210
Yulara	GHI (native)	2950	0	0	-12.872	1.458	970.905	1391.457
NIST	AC meter power (kW)	2127	0	0	-0.647	0.000	199.400	258.400
NIST	Temperature (°C)	2127	0	0	-12.820	15.300	29.260	38.850
NIST	GHI (W/m^2)	2147	0	0	-18.771	-5.703	787.316	1412.584

Table S12: Parameter counts for the two input widths. Changed projections follow from the input width; they are not new models or transferred checkpoints.

Implementation	17 channels	7 channels	Difference
Discrete recurrent	99,362	96,562	-2,800
Inverted-variate	194,960	102,800	-92,160
Joint-patch	148,112	140,432	-7,680
Depthwise TCN	683,024	682,384	-640

7 External preprocessing and analysis details

The seven channels, in order, are historical AC power, temperature, GHI, their three original missingness masks, and the Isolation Forest flag. Non-numeric and non-finite inputs are missing. KNN imputation uses five neighbors and is fitted on Train numerical inputs. Isolation Forest uses the imputed Train values (100 trees, contamination 0.01, random state 42). Masks and flag are appended before the seven-channel MinMaxScaler is fitted on Train. The target scaler uses finite unimputed Train power. Validation and Test only transform; labels are never imputed. Finite negative external targets are retained. Alice's frozen nonnegative-target rule is unchanged.

Train is January–August 2017, Validation September, and Test October–December, interpreted in each fixed coordinate. Windows retain the regular grid, stay within one split, contain 72 historical positions, and start their targets at origin+5 min. Training and Validation use full H144 targets and true finite origin power. Test eligibility is evaluated separately at each prefix. The 1% daylight threshold uses the site's Train maximum and future true labels only for evaluation. Daily uses an exact timestamp-minus-24-hour join, with the same elementwise intersection for every model, seed, and deterministic baseline. Common points do not imply common historical information.

The external primary model is Inverted-variate, selected before external Test predictions. All four implementations and every seed remain in the secondary tables. RMSE skills use matched reference errors and no epsilon when a reference is zero. Bias is mean prediction minus observation; negative R^2 is retained and zero target variance leaves it undefined. Sample SD uses three seeds with divisor two. Deterministic baselines have no stochastic seed repetitions. The external Test split was inspected for data quality, and is a held-out performance split rather than untouched external confirmation.

8 External quantitative evidence

The tables below are generated from `DATA_AUDIT_SUMMARY.csv`, `metrics_per_seed.csv`, and `metrics_summary_mean_sd.csv`. The complete per-seed CSV includes RMSE, MAE, bias, R^2, Train-range nRMSE, both matched skills, origin/target counts, and run metadata. The complete summary CSV includes means and sample SDs for all metrics. They are repository materials; the compact PDF tables preserve all model/seed/horizon/scope combinations without printing the entire long-form data inventory.

Table S13: All 96 external support groups, compressed to 24 rows. Each cell gives origins / valid target points. P: primary; D: Daily-matched. Train/Validation prefix counts are descriptive eligibility inventories; fitting uses H144 only.

Site	Split	H	P full	P daylight	D full	D daylight
Yulara	train	H12	69,861 / 838,332	35,109 / 388,983	69,656 / 835,770	34,948 / 387,108
Yulara	train	H48	69,717 / 3,346,416	43,727 / 1,551,419	69,548 / 3,337,032	43,566 / 1,544,738
Yulara	train	H96	69,525 / 6,674,400	55,170 / 3,091,616	69,404 / 6,657,936	55,009 / 3,080,543

Site	Split	H	P full	P daylight	D full	D daylight
Yulara	train	H144	69,333 / 9,983,952	66,248 / 4,623,382	69,260 / 9,962,712	66,098 / 4,610,221
Yulara	validation	H12	8,459 / 101,508	4,329 / 47,952	8,411 / 100,776	4,281 / 47,232
Yulara	validation	H48	8,315 / 399,120	5,305 / 189,051	8,303 / 396,198	5,293 / 186,177
Yulara	validation	H96	8,135 / 780,960	6,556 / 371,232	8,135 / 776,430	6,556 / 366,798
Yulara	validation	H144	8,037 / 1,157,328	7,802 / 547,225	8,037 / 1,152,674	7,802 / 542,715
Yulara	test	H12	26,412 / 316,944	14,784 / 164,664	26,412 / 316,944	14,784 / 164,664
Yulara	test	H48	26,376 / 1,266,048	18,063 / 658,095	26,376 / 1,266,048	18,063 / 658,095
Yulara	test	H96	26,328 / 2,527,488	22,393 / 1,313,250	26,328 / 2,527,488	22,393 / 1,313,250
Yulara	test	H144	26,280 / 3,784,320	26,192 / 1,963,842	26,280 / 3,784,320	26,192 / 1,963,842
NIST	train	H12	68,837 / 826,044	34,613 / 380,214	68,615 / 821,094	34,479 / 376,830
NIST	train	H48	66,958 / 3,213,984	41,646 / 1,457,504	66,789 / 3,196,711	41,510 / 1,445,846
NIST	train	H96	64,717 / 6,212,832	50,964 / 2,790,328	64,596 / 6,183,604	50,828 / 2,771,505
NIST	train	H144	62,643 / 9,020,592	59,066 / 4,062,345	62,570 / 8,982,982	58,940 / 4,039,615
NIST	validation	H12	8,290 / 99,480	4,132 / 45,799	8,290 / 99,240	4,132 / 45,655
NIST	validation	H48	7,543 / 362,064	4,548 / 159,943	7,543 / 361,151	4,548 / 159,368
NIST	validation	H96	6,690 / 642,240	5,099 / 274,705	6,690 / 640,575	5,099 / 273,662
NIST	validation	H144	5,915 / 851,760	5,625 / 364,699	5,915 / 849,433	5,625 / 363,282
NIST	test	H12	25,219 / 302,628	10,161 / 109,258	25,219 / 300,173	10,158 / 108,291
NIST	test	H48	24,240 / 1,163,520	12,577 / 413,664	24,240 / 1,154,170	12,574 / 410,266
NIST	test	H96	23,037 / 2,211,552	15,700 / 775,182	23,037 / 2,194,678	15,697 / 769,372
NIST	test	H144	21,885 / 3,151,440	18,602 / 1,106,213	21,885 / 3,128,676	18,599 / 1,098,178

Table S14: Primary external mean/sample-SD errors (kW), mean bias (kW), and mean R^2. Complete per-seed values and all metric SDs are in the accompanying CSV.

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	H12	full	Discrete recurrent	12.320 ± 0.369	5.896 ± 0.254	-0.718	0.871
Yulara	H12	daylight	Discrete recurrent	16.672 ± 1.060	10.086 ± 0.699	-1.920	0.729
Yulara	H48	full	Discrete recurrent	15.816 ± 0.201	8.184 ± 0.184	-1.600	0.788
Yulara	H48	daylight	Discrete recurrent	21.271 ± 0.759	13.773 ± 0.686	-4.459	0.560
Yulara	H96	full	Discrete recurrent	17.866 ± 0.456	9.606 ± 0.284	-2.948	0.729
Yulara	H96	daylight	Discrete recurrent	24.289 ± 0.859	16.994 ± 0.694	-6.652	0.427
Yulara	H144	full	Discrete recurrent	17.883 ± 0.759	9.740 ± 0.381	-3.381	0.728
Yulara	H144	daylight	Discrete recurrent	24.460 ± 1.206	17.601 ± 0.812	-7.243	0.419
Yulara	H12	full	Inverted-variate	10.289 ± 0.032	5.888 ± 0.282	-0.290	0.910
Yulara	H12	daylight	Inverted-variate	13.988 ± 0.063	9.455 ± 0.326	-0.920	0.810
Yulara	H48	full	Inverted-variate	13.368 ± 0.140	8.360 ± 0.147	-0.668	0.848
Yulara	H48	daylight	Inverted-variate	17.659 ± 0.242	12.870 ± 0.031	-3.286	0.697
Yulara	H96	full	Inverted-variate	15.699 ± 0.056	10.028 ± 0.067	-1.585	0.791
Yulara	H96	daylight	Inverted-variate	20.927 ± 0.150	16.078 ± 0.033	-4.723	0.575
Yulara	H144	full	Inverted-variate	15.805 ± 0.155	10.226 ± 0.084	-2.132	0.788
Yulara	H144	daylight	Inverted-variate	21.203 ± 0.221	16.519 ± 0.312	-5.435	0.564
Yulara	H12	full	Joint-patch	12.598 ± 0.259	7.221 ± 0.656	0.574	0.865
Yulara	H12	daylight	Joint-patch	16.760 ± 0.667	10.831 ± 0.416	-0.461	0.727
Yulara	H48	full	Joint-patch	14.633 ± 0.085	9.118 ± 0.064	0.129	0.818
Yulara	H48	daylight	Joint-patch	18.733 ± 0.338	13.176 ± 0.194	-2.683	0.659
Yulara	H96	full	Joint-patch	16.795 ± 0.338	10.695 ± 0.299	-0.501	0.761
Yulara	H96	daylight	Joint-patch	21.426 ± 0.282	16.043 ± 0.313	-4.156	0.554
Yulara	H144	full	Joint-patch	17.012 ± 0.389	10.728 ± 0.381	-1.423	0.754
Yulara	H144	daylight	Joint-patch	22.151 ± 0.361	16.695 ± 0.477	-5.308	0.524
Yulara	H12	full	Depthwise TCN	10.562 ± 0.929	5.916 ± 0.902	1.151	0.905
Yulara	H12	daylight	Depthwise TCN	14.035 ± 0.912	8.941 ± 0.968	1.103	0.808
Yulara	H48	full	Depthwise TCN	15.383 ± 1.228	10.516 ± 1.150	-0.608	0.798

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	H48	daylight	Depthwise TCN	19.096 ± 1.564	14.277 ± 1.444	-3.816	0.644
Yulara	H96	full	Depthwise TCN	18.435 ± 1.422	12.851 ± 1.196	-2.877	0.710
Yulara	H96	daylight	Depthwise TCN	23.493 ± 1.947	18.484 ± 1.664	-7.385	0.462
Yulara	H144	full	Depthwise TCN	18.875 ± 1.798	13.265 ± 1.285	-4.054	0.696
Yulara	H144	daylight	Depthwise TCN	24.227 ± 2.457	19.261 ± 1.895	-8.424	0.427
NIST	H12	full	Discrete recurrent	17.303 ± 0.462	8.521 ± 0.437	-0.530	0.914
NIST	H12	daylight	Discrete recurrent	28.431 ± 0.837	19.949 ± 1.509	-0.886	0.815
NIST	H48	full	Discrete recurrent	27.667 ± 0.249	15.150 ± 0.439	1.025	0.776
NIST	H48	daylight	Discrete recurrent	43.523 ± 0.487	31.982 ± 0.658	-5.539	0.566
NIST	H96	full	Discrete recurrent	39.847 ± 1.254	22.530 ± 0.713	7.092	0.522
NIST	H96	daylight	Discrete recurrent	55.504 ± 0.447	42.326 ± 0.203	-0.255	0.286
NIST	H144	full	Discrete recurrent	44.383 ± 1.235	25.486 ± 0.555	12.007	0.402
NIST	H144	daylight	Discrete recurrent	61.477 ± 0.774	48.059 ± 0.379	10.867	0.116
NIST	H12	full	Inverted-variate	17.822 ± 0.113	10.260 ± 0.350	-2.118	0.908
NIST	H12	daylight	Inverted-variate	28.404 ± 0.093	20.126 ± 0.295	-1.457	0.816
NIST	H48	full	Inverted-variate	28.504 ± 0.247	17.015 ± 0.635	-0.128	0.762
NIST	H48	daylight	Inverted-variate	44.434 ± 0.994	32.967 ± 0.723	-5.270	0.548
NIST	H96	full	Inverted-variate	40.216 ± 0.741	25.093 ± 0.771	7.299	0.513
NIST	H96	daylight	Inverted-variate	55.675 ± 0.781	43.249 ± 0.454	1.708	0.281
NIST	H144	full	Inverted-variate	44.617 ± 0.684	28.558 ± 0.456	12.668	0.396
NIST	H144	daylight	Inverted-variate	60.610 ± 1.033	48.410 ± 0.699	12.595	0.141
NIST	H12	full	Joint-patch	19.415 ± 0.415	11.401 ± 0.656	-2.959	0.891
NIST	H12	daylight	Joint-patch	31.167 ± 0.581	23.467 ± 0.262	-6.610	0.778
NIST	H48	full	Joint-patch	29.529 ± 0.419	18.444 ± 0.354	0.592	0.745
NIST	H48	daylight	Joint-patch	45.040 ± 1.037	34.363 ± 0.724	-7.829	0.535
NIST	H96	full	Joint-patch	41.754 ± 0.718	26.312 ± 0.840	9.517	0.475
NIST	H96	daylight	Joint-patch	55.823 ± 0.518	43.843 ± 0.020	3.042	0.277
NIST	H144	full	Joint-patch	46.485 ± 0.921	29.300 ± 0.777	14.868	0.344
NIST	H144	daylight	Joint-patch	62.685 ± 0.781	49.975 ± 0.634	15.056	0.081
NIST	H12	full	Depthwise TCN	17.736 ± 0.079	9.944 ± 0.311	-0.194	0.909
NIST	H12	daylight	Depthwise TCN	28.426 ± 0.152	20.542 ± 0.362	-3.481	0.815
NIST	H48	full	Depthwise TCN	29.116 ± 0.778	19.250 ± 0.882	3.483	0.752
NIST	H48	daylight	Depthwise TCN	42.774 ± 0.129	32.408 ± 0.163	-4.493	0.581
NIST	H96	full	Depthwise TCN	41.159 ± 1.806	27.886 ± 1.548	10.772	0.490
NIST	H96	daylight	Depthwise TCN	54.655 ± 0.514	43.028 ± 0.536	3.178	0.307
NIST	H144	full	Depthwise TCN	45.286 ± 2.493	30.952 ± 1.858	15.401	0.376
NIST	H144	daylight	Depthwise TCN	60.163 ± 1.805	48.525 ± 1.404	12.876	0.153

Table S15: Daily-matched external mean/sample-SD errors (kW), mean bias (kW), and mean R^2. Complete per-seed values and all metric SDs are in the accompanying CSV.

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	H12	full	Discrete recurrent	12.320 ± 0.369	5.896 ± 0.254	-0.718	0.871
Yulara	H12	daylight	Discrete recurrent	16.672 ± 1.060	10.086 ± 0.699	-1.920	0.729
Yulara	H48	full	Discrete recurrent	15.816 ± 0.201	8.184 ± 0.184	-1.600	0.788
Yulara	H48	daylight	Discrete recurrent	21.271 ± 0.759	13.773 ± 0.686	-4.459	0.560
Yulara	H96	full	Discrete recurrent	17.866 ± 0.456	9.606 ± 0.284	-2.948	0.729
Yulara	H96	daylight	Discrete recurrent	24.289 ± 0.859	16.994 ± 0.694	-6.652	0.427
Yulara	H144	full	Discrete recurrent	17.883 ± 0.759	9.740 ± 0.381	-3.381	0.728
Yulara	H144	daylight	Discrete recurrent	24.460 ± 1.206	17.601 ± 0.812	-7.243	0.419
Yulara	H12	full	Inverted-variate	10.289 ± 0.032	5.888 ± 0.282	-0.290	0.910
Yulara	H12	daylight	Inverted-variate	13.988 ± 0.063	9.455 ± 0.326	-0.920	0.810
Yulara	H48	full	Inverted-variate	13.368 ± 0.140	8.360 ± 0.147	-0.668	0.848

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	H48	daylight	Inverted-variate	17.659 ± 0.242	12.870 ± 0.031	-3.286	0.697
Yulara	H96	full	Inverted-variate	15.699 ± 0.056	10.028 ± 0.067	-1.585	0.791
Yulara	H96	daylight	Inverted-variate	20.927 ± 0.150	16.078 ± 0.033	-4.723	0.575
Yulara	H144	full	Inverted-variate	15.805 ± 0.155	10.226 ± 0.084	-2.132	0.788
Yulara	H144	daylight	Inverted-variate	21.203 ± 0.221	16.519 ± 0.312	-5.435	0.564
Yulara	H12	full	Joint-patch	12.598 ± 0.259	7.221 ± 0.656	0.574	0.865
Yulara	H12	daylight	Joint-patch	16.760 ± 0.667	10.831 ± 0.416	-0.461	0.727
Yulara	H48	full	Joint-patch	14.633 ± 0.085	9.118 ± 0.064	0.129	0.818
Yulara	H48	daylight	Joint-patch	18.733 ± 0.338	13.176 ± 0.194	-2.683	0.659
Yulara	H96	full	Joint-patch	16.795 ± 0.338	10.695 ± 0.299	-0.501	0.761
Yulara	H96	daylight	Joint-patch	21.426 ± 0.282	16.043 ± 0.313	-4.156	0.554
Yulara	H144	full	Joint-patch	17.012 ± 0.389	10.728 ± 0.381	-1.423	0.754
Yulara	H144	daylight	Joint-patch	22.151 ± 0.361	16.695 ± 0.477	-5.308	0.524
Yulara	H12	full	Depthwise TCN	10.562 ± 0.929	5.916 ± 0.902	1.151	0.905
Yulara	H12	daylight	Depthwise TCN	14.035 ± 0.912	8.941 ± 0.968	1.103	0.808
Yulara	H48	full	Depthwise TCN	15.383 ± 1.228	10.516 ± 1.150	-0.608	0.798
Yulara	H48	daylight	Depthwise TCN	19.096 ± 1.564	14.277 ± 1.444	-3.816	0.644
Yulara	H96	full	Depthwise TCN	18.435 ± 1.422	12.851 ± 1.196	-2.877	0.710
Yulara	H96	daylight	Depthwise TCN	23.493 ± 1.947	18.484 ± 1.664	-7.385	0.462
Yulara	H144	full	Depthwise TCN	18.875 ± 1.798	13.265 ± 1.285	-4.054	0.696
Yulara	H144	daylight	Depthwise TCN	24.227 ± 2.457	19.261 ± 1.895	-8.424	0.427
NIST	H12	full	Discrete recurrent	17.285 ± 0.460	8.518 ± 0.436	-0.506	0.914
NIST	H12	daylight	Discrete recurrent	28.412 ± 0.831	19.953 ± 1.511	-0.814	0.815
NIST	H48	full	Discrete recurrent	27.662 ± 0.242	15.155 ± 0.439	1.079	0.776
NIST	H48	daylight	Discrete recurrent	43.507 ± 0.504	31.985 ± 0.666	-5.402	0.566
NIST	H96	full	Discrete recurrent	39.850 ± 1.259	22.536 ± 0.715	7.153	0.522
NIST	H96	daylight	Discrete recurrent	55.469 ± 0.448	42.304 ± 0.208	-0.115	0.285
NIST	H144	full	Discrete recurrent	44.388 ± 1.245	25.484 ± 0.560	12.068	0.401
NIST	H144	daylight	Discrete recurrent	61.445 ± 0.780	48.019 ± 0.384	11.001	0.115
NIST	H12	full	Inverted-variate	17.785 ± 0.116	10.243 ± 0.354	-2.114	0.909
NIST	H12	daylight	Inverted-variate	28.364 ± 0.090	20.106 ± 0.289	-1.412	0.816
NIST	H48	full	Inverted-variate	28.493 ± 0.242	17.008 ± 0.640	-0.092	0.762
NIST	H48	daylight	Inverted-variate	44.420 ± 0.982	32.960 ± 0.707	-5.152	0.547
NIST	H96	full	Inverted-variate	40.207 ± 0.751	25.096 ± 0.779	7.373	0.513
NIST	H96	daylight	Inverted-variate	55.612 ± 0.784	43.213 ± 0.458	1.883	0.281
NIST	H144	full	Inverted-variate	44.609 ± 0.690	28.550 ± 0.462	12.745	0.395
NIST	H144	daylight	Inverted-variate	60.544 ± 1.041	48.352 ± 0.704	12.775	0.141
NIST	H12	full	Joint-patch	19.393 ± 0.415	11.397 ± 0.660	-2.942	0.891
NIST	H12	daylight	Joint-patch	31.142 ± 0.583	23.461 ± 0.262	-6.554	0.778
NIST	H48	full	Joint-patch	29.536 ± 0.426	18.453 ± 0.362	0.628	0.745
NIST	H48	daylight	Joint-patch	45.048 ± 1.027	34.383 ± 0.719	-7.722	0.534
NIST	H96	full	Joint-patch	41.763 ± 0.726	26.320 ± 0.848	9.578	0.475
NIST	H96	daylight	Joint-patch	55.792 ± 0.508	43.822 ± 0.018	3.186	0.277
NIST	H144	full	Joint-patch	46.495 ± 0.924	29.300 ± 0.780	14.934	0.343
NIST	H144	daylight	Joint-patch	62.653 ± 0.785	49.933 ± 0.638	15.204	0.080
NIST	H12	full	Depthwise TCN	17.715 ± 0.080	9.941 ± 0.315	-0.189	0.909
NIST	H12	daylight	Depthwise TCN	28.396 ± 0.153	20.534 ± 0.369	-3.465	0.816
NIST	H48	full	Depthwise TCN	29.110 ± 0.785	19.262 ± 0.891	3.525	0.752
NIST	H48	daylight	Depthwise TCN	42.745 ± 0.139	32.398 ± 0.174	-4.396	0.581
NIST	H96	full	Depthwise TCN	41.155 ± 1.819	27.899 ± 1.559	10.841	0.489
NIST	H96	daylight	Depthwise TCN	54.599 ± 0.526	42.991 ± 0.543	3.309	0.307
NIST	H144	full	Depthwise TCN	45.279 ± 2.509	30.953 ± 1.869	15.466	0.375
NIST	H144	daylight	Depthwise TCN	60.100 ± 1.824	48.465 ± 1.414	13.007	0.153

Table S16: External primary RMSE (kW) for every seed. The repository metric tables additionally provide every per-seed MAE, bias, R^2, nRMSE and matched skill.

Site	Model	H	Scope	Seed 42	Seed 43	Seed 44
Yulara	Discrete recurrent	H12	full	12.484	11.898	12.579
Yulara	Discrete recurrent	H12	daylight	17.212	15.450	17.353
Yulara	Discrete recurrent	H48	full	15.932	15.583	15.932
Yulara	Discrete recurrent	H48	daylight	21.662	20.396	21.755
Yulara	Discrete recurrent	H96	full	18.205	17.348	18.044
Yulara	Discrete recurrent	H96	daylight	24.878	23.304	24.686
Yulara	Discrete recurrent	H144	full	18.479	17.028	18.140
Yulara	Discrete recurrent	H144	daylight	25.369	23.092	24.920
Yulara	Inverted-variate	H12	full	10.321	10.257	10.290
Yulara	Inverted-variate	H12	daylight	13.998	13.921	14.045
Yulara	Inverted-variate	H48	full	13.295	13.279	13.529
Yulara	Inverted-variate	H48	daylight	17.386	17.745	17.846
Yulara	Inverted-variate	H96	full	15.666	15.668	15.763
Yulara	Inverted-variate	H96	daylight	20.753	21.018	21.008
Yulara	Inverted-variate	H144	full	15.983	15.735	15.697
Yulara	Inverted-variate	H144	daylight	21.429	21.190	20.988
Yulara	Joint-patch	H12	full	12.868	12.575	12.352
Yulara	Joint-patch	H12	daylight	17.427	16.759	16.093
Yulara	Joint-patch	H48	full	14.707	14.540	14.653
Yulara	Joint-patch	H48	daylight	18.964	18.890	18.345
Yulara	Joint-patch	H96	full	17.143	16.469	16.774
Yulara	Joint-patch	H96	daylight	21.752	21.263	21.263
Yulara	Joint-patch	H144	full	17.356	16.590	17.089
Yulara	Joint-patch	H144	daylight	22.454	21.752	22.248
Yulara	Depthwise TCN	H12	full	11.632	10.090	9.963
Yulara	Depthwise TCN	H12	daylight	15.088	13.474	13.543
Yulara	Depthwise TCN	H48	full	16.794	14.797	14.557
Yulara	Depthwise TCN	H48	daylight	20.876	17.941	18.471
Yulara	Depthwise TCN	H96	full	20.075	17.672	17.558
Yulara	Depthwise TCN	H96	daylight	25.697	22.002	22.781
Yulara	Depthwise TCN	H144	full	20.950	17.914	17.761
Yulara	Depthwise TCN	H144	daylight	27.037	22.480	23.164
NIST	Discrete recurrent	H12	full	17.828	16.957	17.124
NIST	Discrete recurrent	H12	daylight	29.355	27.725	28.213
NIST	Discrete recurrent	H48	full	27.767	27.383	27.851
NIST	Discrete recurrent	H48	daylight	43.298	44.081	43.189
NIST	Discrete recurrent	H96	full	40.001	38.524	41.017
NIST	Discrete recurrent	H96	daylight	55.020	55.899	55.592
NIST	Discrete recurrent	H144	full	44.185	43.258	45.705
NIST	Discrete recurrent	H144	daylight	61.184	60.893	62.355
NIST	Inverted-variate	H12	full	17.910	17.862	17.695
NIST	Inverted-variate	H12	daylight	28.415	28.492	28.306
NIST	Inverted-variate	H48	full	28.594	28.695	28.225
NIST	Inverted-variate	H48	daylight	43.395	45.375	44.534
NIST	Inverted-variate	H96	full	40.813	39.387	40.447
NIST	Inverted-variate	H96	daylight	54.949	55.576	56.501
NIST	Inverted-variate	H144	full	44.558	43.965	45.329
NIST	Inverted-variate	H144	daylight	59.779	60.284	61.767
NIST	Joint-patch	H12	full	19.421	19.827	18.997
NIST	Joint-patch	H12	daylight	30.787	31.836	30.879
NIST	Joint-patch	H48	full	29.045	29.762	29.780
NIST	Joint-patch	H48	daylight	44.920	46.132	44.067

Site	Model	H	Scope	Seed 42	Seed 43	Seed 44
NIST	Joint-patch	H96	full	41.149	41.565	42.548
NIST	Joint-patch	H96	daylight	56.207	56.028	55.233
NIST	Joint-patch	H144	full	46.730	45.467	47.259
NIST	Joint-patch	H144	daylight	63.489	61.930	62.635
NIST	Depthwise TCN	H12	full	17.765	17.797	17.646
NIST	Depthwise TCN	H12	daylight	28.343	28.333	28.601
NIST	Depthwise TCN	H48	full	28.778	30.005	28.563
NIST	Depthwise TCN	H48	daylight	42.795	42.891	42.637
NIST	Depthwise TCN	H96	full	41.116	42.985	39.375
NIST	Depthwise TCN	H96	daylight	54.724	55.132	54.110
NIST	Depthwise TCN	H144	full	45.901	47.414	42.542
NIST	Depthwise TCN	H144	daylight	60.960	61.433	58.096

Table S17: Frozen 24-run matrix, parameter counts, selected Validation epoch/MSE and training seconds. These are historical run records; no training is performed for manuscript preparation.

Site	Model	Seed	Parameters	Best epoch	Val. MSE	Seconds
Yulara	Discrete recurrent	42	96562	8	0.042345	1394.8
Yulara	Discrete recurrent	43	96562	13	0.029118	2175.8
Yulara	Discrete recurrent	44	96562	6	0.042264	1152.9
Yulara	Inverted-variate	42	102800	20	0.015972	198.4
Yulara	Inverted-variate	43	102800	11	0.016846	128.2
Yulara	Inverted-variate	44	102800	10	0.017832	120.1
Yulara	Joint-patch	42	140432	7	0.031496	96.3
Yulara	Joint-patch	43	140432	12	0.027561	138.2
Yulara	Joint-patch	44	140432	8	0.031563	103.3
Yulara	Depthwise TCN	42	682384	1	0.046002	47.9
Yulara	Depthwise TCN	43	682384	10	0.028054	126.4
Yulara	Depthwise TCN	44	682384	9	0.021205	117.3
NIST	Discrete recurrent	42	96562	16	0.016202	1850.3
NIST	Discrete recurrent	43	96562	11	0.016805	1404.9
NIST	Discrete recurrent	44	96562	15	0.016085	2360.2
NIST	Inverted-variate	42	102800	19	0.017901	171.1
NIST	Inverted-variate	43	102800	16	0.018544	150.3
NIST	Inverted-variate	44	102800	24	0.017946	179.4
NIST	Joint-patch	42	140432	24	0.017498	179.5
NIST	Joint-patch	43	140432	21	0.017589	178.6
NIST	Joint-patch	44	140432	18	0.017219	164.6
NIST	Depthwise TCN	42	682384	25	0.017715	184.4
NIST	Depthwise TCN	43	682384	25	0.018257	181.2
NIST	Depthwise TCN	44	682384	25	0.018794	181.9

Table S18: External post hoc descriptive envelope against Daily on identical target points. It is not the prespecified primary model or a deployable strategy.

Site	H	Scope	Envelope member	RMSE	Daily	Skill (%)
Yulara	H12	full	Inverted-variate	10.289	16.147	+36.28
Yulara	H12	daylight	Inverted-variate	13.988	22.302	+37.28
Yulara	H48	full	Inverted-variate	13.368	16.154	+17.25
Yulara	H48	daylight	Inverted-variate	17.659	22.305	+20.83

Site	H	Scope	Envelope member	RMSE	Daily	Skill (%)
Yulara	H96	full	Inverted-variate	15.699	16.137	+2.71
Yulara	H96	daylight	Inverted-variate	20.927	22.286	+6.10
Yulara	H144	full	Inverted-variate	15.805	16.066	+1.62
Yulara	H144	daylight	Inverted-variate	21.203	22.201	+4.50
NIST Ground	H12	full	Discrete recurrent	17.285	42.074	+58.92
NIST Ground	H12	daylight	Inverted-variate	28.364	69.198	+59.01
NIST Ground	H48	full	Discrete recurrent	27.662	41.955	+34.07
NIST Ground	H48	daylight	Depthwise TCN	42.745	69.477	+38.48
NIST Ground	H96	full	Discrete recurrent	39.850	41.831	+4.74
NIST Ground	H96	daylight	Depthwise TCN	54.599	69.706	+21.67
NIST Ground	H144	full	Discrete recurrent	44.388	41.899	-5.94
NIST Ground	H144	daylight	Depthwise TCN	60.100	69.748	+13.83

9 Evidence verification for the external replications

All 24 completed runs have seven-channel checkpoints and finite saved predictions. Before manuscript editing, the M1-R data/protocol tests passed 43/43, M2 ordinary tests 17/17, and artifact tests 10/10, with zero failures, errors, or skips. The independent verifier recomputed 10,432 numerical comparisons from saved artifacts without importing the training metric functions. All passed. Checkpoints reproduced saved predictions within the registered numerical tolerance, and the original 96 support groups remained unchanged.

The original 36 Alice checkpoints, completion records, and prediction files were also inspected as 36 complete, correctly identified groups. Manuscript preparation performs no training or checkpoint updates. External source data and both sets of model artifacts remain read-only, with file sizes and nanosecond modification times checked before and after. Original Alice raw files were not accessed during this manuscript revision. Programmatic paper comparisons distinguish Alice's descriptive focal model from the external prespecified model and from the post hoc envelope; no pooled forty-comparison inference is made.

10 Provider documentation

Official sources checked on September 7, 2026 include the NIST data paper (doi:10.6028/jres.122.040), dataset (doi:10.18434/M3S67G), NIST Data Dictionary v1.0, the Data.gov NIST campus PV catalogue, the DKA Yulara download/system pages, glossary, and download terms. The provider URLs are included in the main bibliography and the frozen external protocol. The missing 2017 log and partial Yulara metadata remain explicitly recorded; unavailable documents are not reconstructed from observed model behavior.

11 Alice rank sensitivity

Figure S1 gives all Alice primary ranks. Figures S2 and S3 retain the horizon/scope curves and the hardware-specific accuracy–latency comparison.

12 Independent evidence verification

An independent NumPy/Pandas verifier reads the saved H144 predictions, labels, timestamps, and validity arrays without importing the production metric or mask functions. It re-derives primary and Daily-matched masks, Persistence forecasts, RMSE, MAE, Train-range nRMSE, ranks, and sample counts. The frozen Alice audit records all 4,414 comparisons passing the registered numerical tolerances; the maximum absolute discrepancy was 8.51×10^{-12} and the maximum relative discrepancy was 3.23×10^{-11}. The verifier confirmed primary wins of 12, 9, 2, and 1 for the Inverted-variate, Depthwise, Joint-patch, and Discrete implementations, respectively, with zero for Last-value Persistence. It also confirmed that Daily Persistence was lower than the post hoc best-of-four neural envelope in 22 comparisons; the two envelope wins were Hanwha H12 full-timeline and daylight.

13 Alice inference measurement details

Parameter counts include all learned tensors. Batch-one latency and batch-256 throughput were measured in evaluation and inference modes on an NVIDIA GeForce RTX 3060 Laptop GPU with float32 inputs of shape $[B, 72, 17]$. Warm-up preceded synchronized repetitions. Timing excludes model loading, disk input/output, data loading, and host-to-device transfer. Peak allocation is reported in MiB. These hardware-specific measurements support only within-environment comparisons.

14 Unmatched-sample diagnostic

Deleting unavailable lag points from Daily Persistence alone changes its target support and can reverse a headline comparison even when each metric is calculated correctly. The matched-support analysis applies one point mask to Daily Persistence, Last-value Persistence, every neural seed, and the labels. Unmatched-support values are therefore treated only as a methodological diagnostic and never as model-performance evidence.

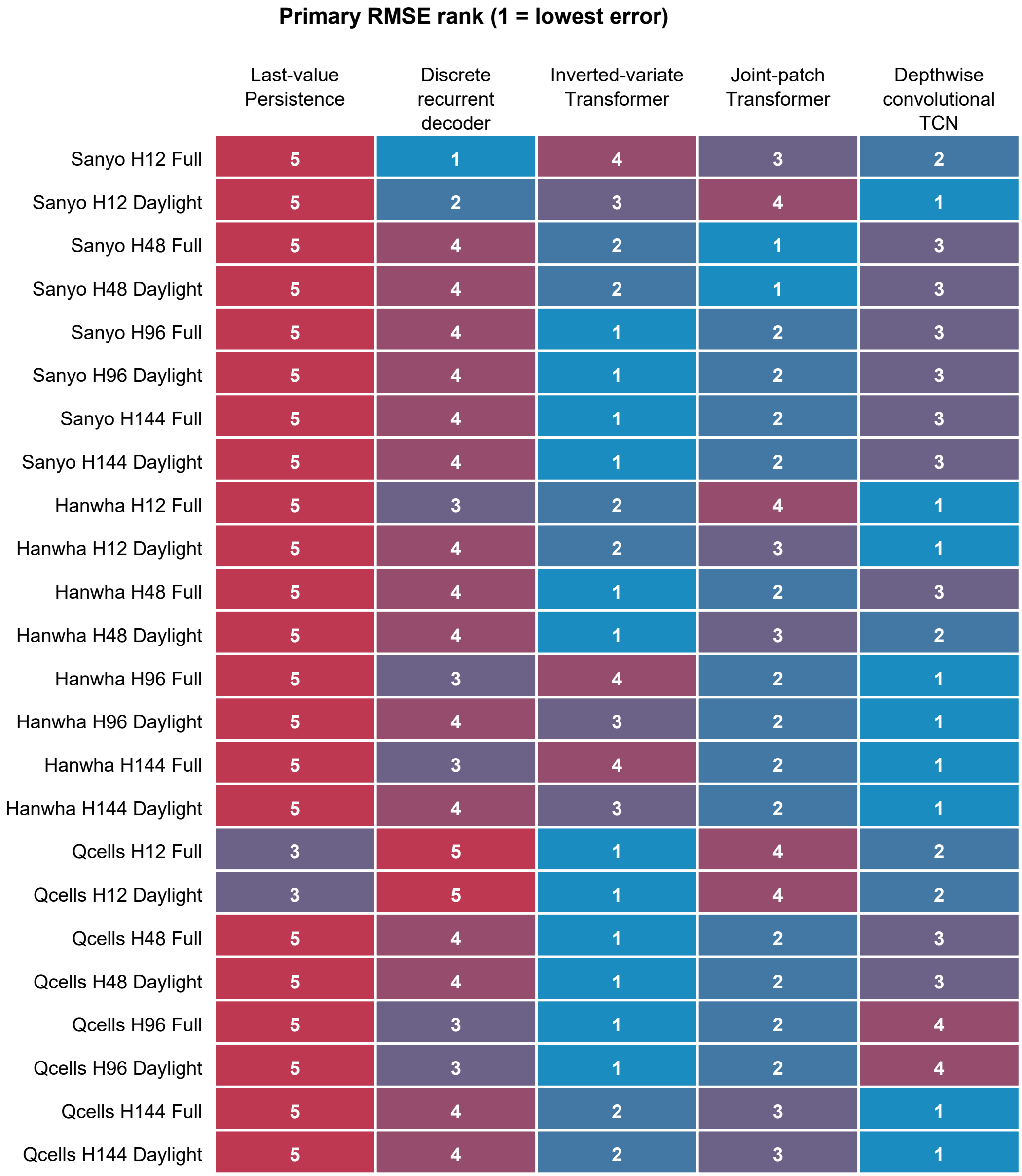

Figure S1: RMSE ranks for all 24 primary array–horizon–scope combinations. Rank one denotes the smallest error. Short column names correspond to Last-value Persistence, Discrete recurrent decoder, Inverted-variate Transformer, Joint-patch Transformer, and Depthwise convolutional TCN.

Alt text: Heat map of RMSE ranks across 24 array, horizon, and scope combinations. Columns denote Last-value Persistence, Discrete recurrent, Inverted-variate, Joint-patch, and Depthwise TCN; rankings change across conditions.

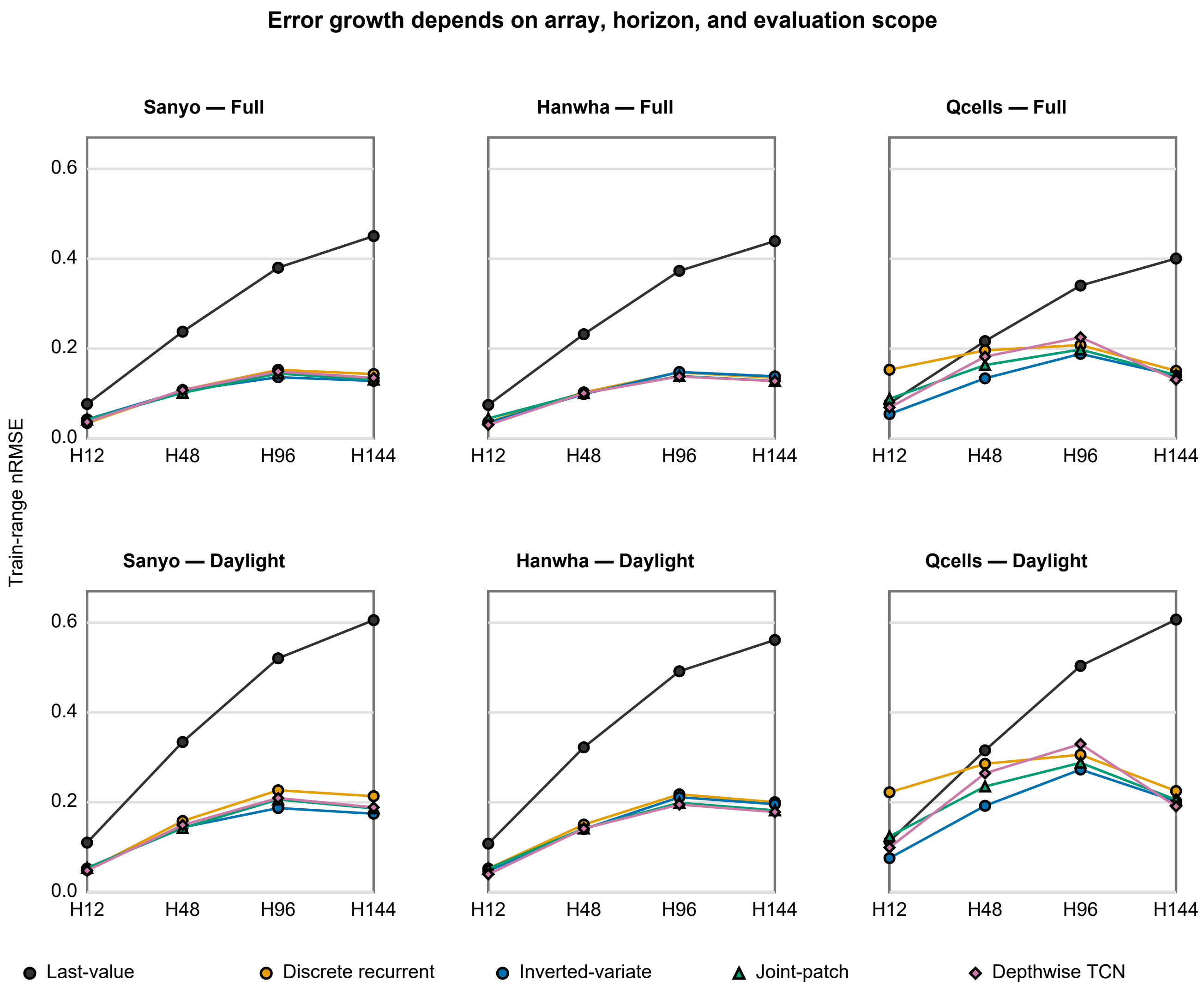

Figure S2: Alice primary Train-range nRMSE for every horizon, array, and scope.
Alt text: Six panels separate full/daylight scopes and three co-located arrays, showing all four compact models and Last-value across H12, H48, H96, and H144.

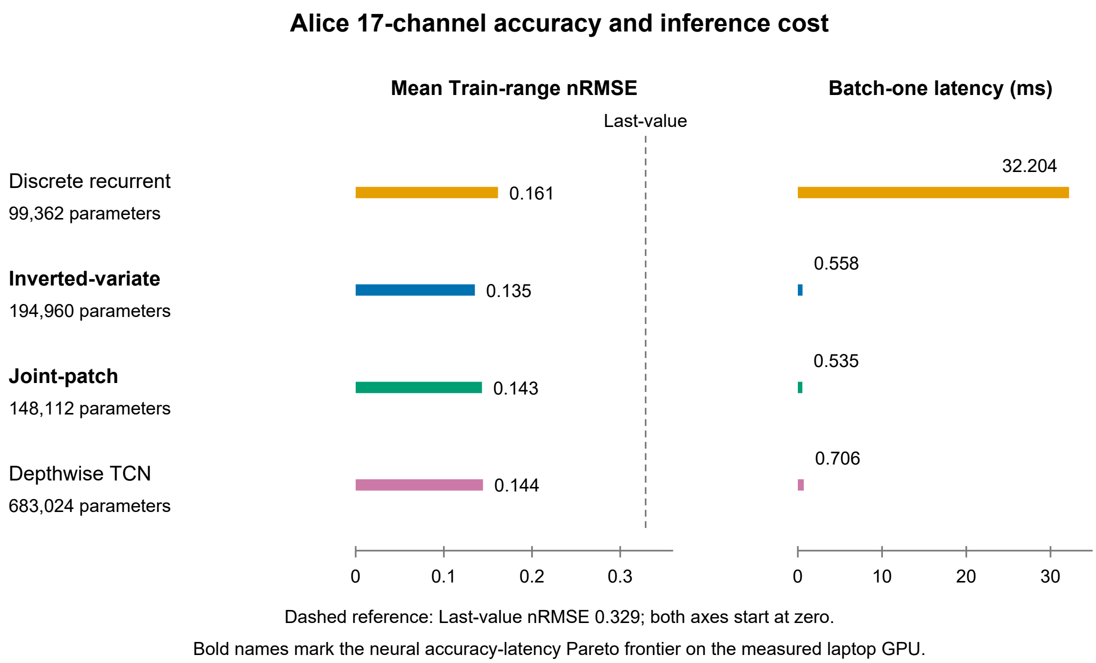

Figure S3: Alice 17-channel accuracy and measured inference latency on the documented RTX 3060 Laptop GPU. Timing excludes data ingestion and preprocessing.

Alt text: Aligned rows show parameters, average normalized error and latency for four implementations on zero-based axes. A dashed line marks Last-value error. Joint-patch is fastest and Inverted-variate has the lowest average normalized error.